ATTNDIFF: Attention-based Differential Fingerprinting for Large Language Models

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Abstract

Protecting the intellectual property of open-weight large language models (LLMs) requires verifying whether a suspect model is derived from a victim model despite common laundering operations such as fine-tuning (including PPO/DPO), pruning/compression, and model merging. We propose ATTNDIFF, a data-efficient **white-box** framework that *extracts* fingerprints from models via intrinsic information-routing behavior. ATTNDIFF probes minimally edited prompt pairs that induce controlled semantic conflicts, captures *differential attention* patterns, summarizes them with compact spectral descriptors, and compares models using CKA. Across Llama-2/3 and Qwen2.5 (3B–14B) and additional open-source families, it yields high similarity for related derivatives while separating unrelated model families (e.g., > 0.98 vs. < 0.22 with $M = 60$ probes). With 5–60 multi-domain probes, it supports practical provenance verification and accountability; our open-source implementation is available at `https://anonymous.4open.science/r/AttnDiff-37A1/`.

1 Introduction

The rapid advancement of Large Language Models (LLMs), such as the Llama series [46], Qwen [3], and DeepSeek [28], has revolutionized natural language understanding and generation. However, the prohibitive costs associated with large-scale data curation and massive computational resources render model weights the core intellectual property (IP) of AI entities. In the current open-weight ecosystem, the risk of unauthorized redistribution, illicit fine-tuning, and model laundering has escalated significantly [57, 67]. Consequently, model provenance verification—the forensic capability to trace a suspect model back to its architectural or parametric origin—has become a cornerstone for copyright enforcement and IP protection [57, 17].

Existing methodologies for model ownership verification are generally bifurcated into invasive and non-invasive (passive) approaches [57]. Invasive techniques, such as backdoor watermark-based fingerprints, embed ownership signals by modifying the training process [56, 51, 25, 35]; however, they necessarily alter model weights, inducing a trade-off between fingerprint capacity and utility and potentially introducing hard-to-characterize vulnerabilities, such as spurious trigger activations or degraded generalization [52, 56, 51]. Moreover, if the model is stolen before embedding, invasive mechanisms do not support retrospective provenance analysis, which is particularly problematic in open-weight, post-hoc forensic settings where released models typically lack pre-embedded protection and thus cannot be reliably traced *ex post*.

In contrast, non-invasive approaches—commonly referred to as *intrinsic* model fingerprinting—extract signatures without altering parameters. Existing intrinsic fingerprints fall into four families with distinct robustness gaps: **parameter-based fingerprints** [66] operate in weight space and are fragile under structural edits such as structured pruning (Sec. 5.4); **representation-based finger-**

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prints [63, 67] summarize internal activations but remain vulnerable to architectural changes and can exhibit distribution shifts under distillation (Appendix F.5) and preference optimization (PO) fine-tuning (Appendix F.1); **adversarial-example-based fingerprints** [21] rely on crafted triggers that can be neutralized by input perturbations [56, 52] and are often high-perplexity or atypical, reducing stealth; and **semantic fingerprints** [37] depend on generated semantics and decoding, making them sensitive to paraphrasing and rewriting attacks (Appendix F.8).

Motivated by these limitations, we propose ATTNDIFF, a robust and data-efficient white-box fingerprinting framework based on *differential attention dynamics* under semantic conflict, i.e., minimally perturbed prompt pairs that invert logical entailment. Rather than relying on static parameters, generic hidden states, or surface-level semantics, ATTNDIFF directly probes how a model re-routes attention when confronted with such prompts. Our central hypothesis is that the resulting information-routing pattern constitutes a stable, intrinsic property of the model. Specifically, this pattern—defined by how self-attention distributes attention weights over tokens encoding logical operators, factual priors, and safety signals—persists under a broad spectrum of model-level attacks. Concretely, ATTNDIFF constructs paired prompts that introduce small lexical pivots, such as single-word substitutions, to flip the underlying entailment (e.g., "parallel lines never intersect" vs. "always intersect"). It then extracts the corresponding differential attention maps across layers and heads, compressing them into compact spectral descriptors. Models are compared in this feature space using centered linear CKA [22], a similarity measure that captures structural alignment between representation spaces and is invariant to isotropic reparameterization. The resulting fingerprints are simultaneously robust to realistic model-level attacks and discriminative across architecturally distinct models.

2 Related work

2.1 Invasive fingerprints

Invasive fingerprints embed ownership signals by modifying models during (pre-)training, typically via weight watermarks or backdoor triggers. Weight watermark-based schemes encode verifiable patterns in parameter space [69, 15, 10, 29, 43] but face capacity–utility trade-offs and may lose robustness under pruning, quantization, or strong fine-tuning [57]. Backdoor-based fingerprints use trigger datasets for black-box attribution [51, 5, 59, 56, 53, 58, 26, 42], yet require training-time control and can be removed by targeted erasure [68]. Recent work explores scalable fingerprint transfer [55, 54], harmless editing [65], and adversarial robustness [49].

2.2 Intrinsic fingerprints

Non-invasive (intrinsic) fingerprints do not inject external signals and instead mine model-inherent properties for provenance verification [57]. Representative families include parameter-space statistics in weight space [66, 17], representation-based descriptors over hidden states/logits [67, 63, 45], semantic output traces [37, 50], and adversarial-example-based triggers [21, 14]. However, these signals can be brittle under realistic transformations, e.g., pruning/architecture edits, distillation, or preference optimization, and/or sensitive to decoding and input perturbations, limiting robustness across fine-tuning, pruning/merging, and distillation.

Among intrinsic approaches, ATTNDIFF departs from parameter-, representation-, and output-level signatures by fingerprinting models through their *differential attention* responses to controlled semantic conflicts, with similarity measured via CKA.

3 Threat model

We study a post-hoc model forensics setting where a *defender* (the legitimate model owner) verifies the provenance of a released or seized *suspect* model, while an *attacker* seeks to evade attribution after illicitly obtaining the model weights. We focus on realistic **white-box** model laundering operations that modify a stolen model's parameters and/or architecture while maintaining utility.

Attacker capabilities. With full white-box access, the attacker can apply arbitrary transformations, including supervised fine-tuning (SFT), alignment and PO (e.g., reinforcement learning from human feedback (RLHF) with Proximal Policy Optimization (PPO) or Direct Preference Optimization

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(DPO)), pruning, quantization, distillation to a different architecture, and model merging or weight interpolation. This setting is consistent with prior work on model fingerprinting [62, 21], where attackers are assumed to have access to the original model and can fine-tune it using validation data to evade attribution.

Defender capabilities. The defender has access to the original reference model and is granted white-box access to the suspect model during verification, consistent with legal/auditing workflows. This allows the defender to inspect internal states and compute fingerprints for both models under the same probing protocol.

4 Design of ATTNDIFF

4.1 Motivation

In post-hoc LLM forensics, a suspect is rarely an "exact-copy": stolen checkpoints are routinely laundered through fine-tuning/alignment, pruning, merging, or distillation. A practical fingerprint should therefore remain stable under such transformations, where weight-space statistics can break under structural edits and hidden-state/logit signatures can drift after strong adaptation.

We encountered an empirical turning point while stress-testing small, highly controlled probes. We constructed minimally edited origin/corrupted prompt pairs that preserve surface form while flipping semantics via a single-word lexical pivot. Under these "controlled semantic conflicts", attention did not drift arbitrarily; models re-allocated attention in a structured manner. More importantly, this re-allocation appeared family-consistent: Llama-2-7B [46] derivatives exhibited closely matched profiles, whereas an unrelated model family such as Qwen2.5-7B [38] followed a systematically different regime. Together, these observations suggested that a model's intrinsic information-routing strategy—how it redistributes self-attention when resolving a semantic contradiction—may encode lineage-specific structure that survives common "laundering" operations. A detailed pilot study quantifying these re-allocation patterns via interpretable structural statistics (concentration, entropy, locality) is presented in Appendix E.

These observations motivate ATTNDIFF: by probing a model with minimally edited prompt pairs that induce controlled semantic conflicts, we can elicit a characteristic re-routing response that is stable within a model lineage yet distinct across unrelated families. The next subsection describes how we operationalize this idea into a practical fingerprinting pipeline.

4.2 ATTNDIFF workflow

Notation. We denote a model with L layers and H attention heads per layer. For a prompt p tokenized into N tokens, $A_{l,h}^{(i)} \in \mathbb{R}^{N\times N}$ represents the post-softmax causal self-attention probability matrix at layer l and head h for the i-th probe pair. We use $\tilde{p}$ and $\tilde{A}$ to denote the corrupted prompt and its corresponding attention map. The differential attention matrix is $\Delta A = \tilde{A} - A$. The fingerprint matrix $F \in \mathbb{R}^{M\times D}$ stacks spectral descriptors from M probe pairs, where $D = LHK$ and K is the spectral rank.

ATTNDIFF fingerprints LLMs by capturing their **differential attention dynamics** under semantic conflict. As shown in Figure 1, the pipeline comprises three stages: **(i) Probe construction**: generate M origin/corrupted prompt pairs via single-word pivot substitutions; **(ii) Fingerprint extraction**: extract differential attention maps ΔA across layers and heads, compress them into spectral descriptors via SVD, and stack into fingerprint matrix $F \in \mathbb{R}^{M\times D}$; **(iii) Similarity measurement**: compare two models by computing centered kernel alignment (CKA) on their fingerprint matrices. Given a victim model and a suspect model, we compute a similarity score $s = \mathrm{CKA}(F, F')$ to quantify their functional similarity. Implementation details are provided in Appendix B.

4.2.1 Probe dataset

We construct a probe set of M prompts spanning six domains: *Code*, *Math*, *Economics*, *Medicine*, *Daily QA*, and *Safe Alignment*. Since post-training typically specializes a model toward a particular domain, we diversify probe domains to reduce domain-specific drift in internal routing behaviors and to provide complementary anchors for fingerprint comparison. In addition, *Safe Alignment* is

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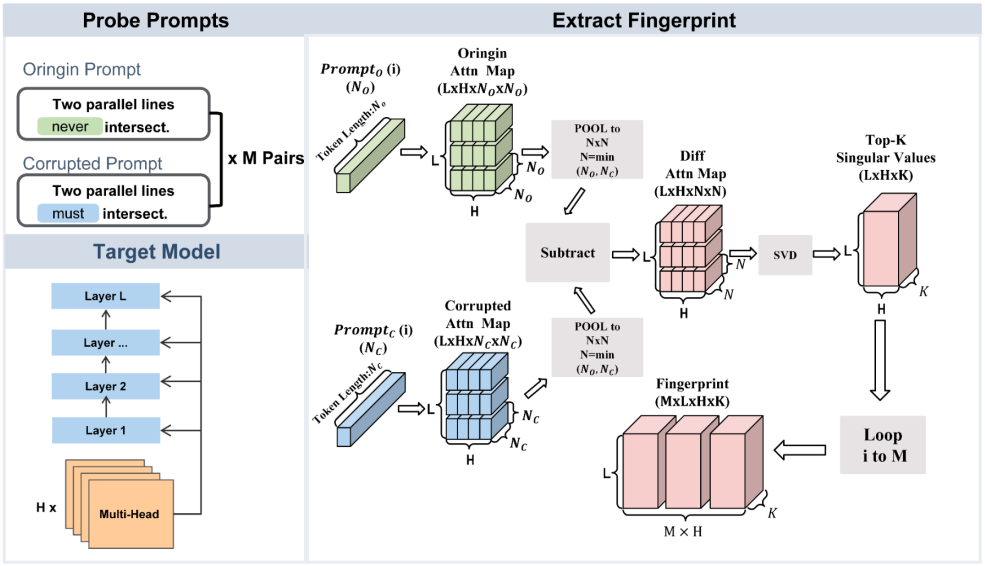


Figure 1: ATTNDIFF pipeline. **Left:** construct M origin/corrupted prompt pairs $(p, \tilde{p})$ via a single-word pivot substitution. **Right:** extract causal attention maps over L layers and H heads, compute $\Delta A = \tilde{A} - A$, and summarize ΔA by SVD with $\texttt{TopK}(\sigma)$ (largest K singular values). Concatenating over heads/layers and stacking over pairs yields the fingerprint matrix F. Parentheses denote dimensions (e.g., N, H, L, M). Differential attention dynamics yield compact, lineage-invariant fingerprints.

included to cover policy-gated behaviors (e.g., refusal or cautious responses) that can follow a distinct internal routing regime compared to capability-oriented queries, providing a complementary anchor under realistic instruction/alignment settings.

We then construct M origin/corrupted probe pairs for fingerprint extraction, as detailed below.

4.2.2 Differential fingerprint

For the i-th probe pair $(p_i, \tilde{p}_i)$, we generate a corrupted prompt $\tilde{p}_i$ from the origin prompt p_i via a *single-word (lexical) pivot* substitution, yielding a *semantic conflict*. Tokenization is performed using the target model's tokenizer on *only the prompt string itself* (excluding chat templates or system prefixes). Typically $\tilde{N} = N$, with rare mismatches due to tokenizer segmentation.

For each layer and head, we extract the post-softmax causal self-attention probability matrices $A^{(i)}_{l,h} \in \mathbb{R}^{N\times N}$ and $\tilde{A}^{(i)}_{l,h} \in \mathbb{R}^{\tilde{N}\times\tilde{N}}$, where strictly upper-triangular entries are masked to zero. We refer to the stacked tensors as the **origin attention map** $\mathcal{A}^{(i)} \in \mathbb{R}^{L\times H\times N\times N}$ and the **corrupted attention map** $\tilde{\mathcal{A}}^{(i)} \in \mathbb{R}^{L\times H\times \tilde{N}\times \tilde{N}}$.

To handle the rare case $\tilde{N} \neq N$, we align attention maps to a common resolution $N^* = \min(N, \tilde{N})$ via **2D adaptive average pooling**, as illustrated in Figure 2. This operator partitions the input matrix into bins and averages within each bin to produce the target resolution, preserving coarse-grained attention distribution without introducing boundary artifacts. We apply this operator to pool each $A^{(i)}_{l,h}$ and $\tilde{A}^{(i)}_{l,h}$ to $\bar{A}^{(i)}_{l,h}, \bar{\tilde{A}}^{(i)}_{l,h} \in \mathbb{R}^{N^*\times N^*}$.

We then compute the **differential attention matrix** $\Delta A^{(i)}_{l,h} = \bar{\tilde{A}}^{(i)}_{l,h} - \bar{A}^{(i)}_{l,h}$ (i.e., $\mathrm{Pool}(\tilde{A}^{(i)}_{l,h}) - \mathrm{Pool}(A^{(i)}_{l,h})$), which highlights attention re-routing induced by semantic conflict under aligned token topology. To obtain a compact, length-invariant signature, we apply truncated SVD to $\Delta A^{(i)}_{l,h}$ and retain the top-K singular values as a spectral descriptor $\mathbf{s}^{(i)}_{l,h} \in \mathbb{R}^K$. In Figure 1, we denote this operation as $\texttt{TopK}(\sigma)$, i.e., selecting the largest K singular values σ returned by SVD. Concatenating

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across all layers and heads yields a one-dimensional instance-level fingerprint vector $\mathbf{x}^{(i)} \in \mathbb{R}^{LHK}$, and stacking over M probes forms the fingerprint matrix $F \in \mathbb{R}^{M \times LHK}$.

4.2.3 Similarity measurement by CKA

To compare two models, we compute fingerprints F and F' on the same probe set and measure similarity using **CKA**. Specifically, we form linear Gram matrices $K = FF^\top$ and $K' = F'F'^\top \in \mathbb{R}^{M \times M}$ that encode inter-probe relational geometry. Let $H = I - \frac{1}{M}\mathbf{1}\mathbf{1}^\top$ be the centering matrix and define centered Gram matrices $\bar{K} = HKH$ and $\bar{K}' = HK'H$. We compute CKA via the Hilbert-Schmidt Independence Criterion (HSIC), where $\mathrm{HSIC}(K, K') \propto \mathrm{tr}(\bar{K}\,\bar{K}')$: $\mathrm{CKA}(F, F') = \frac{\mathrm{HSIC}(K,K')}{\sqrt{\mathrm{HSIC}(K,K)\mathrm{HSIC}(K',K')}}$. We use centered Gram matrices in HSIC to remove mean effects and focus the comparison on similarity structure across probes.

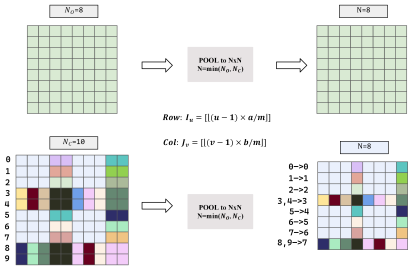


Figure 2: Attention map alignment via 2D adaptive average pooling to resolution $N^* = \min(N, \tilde{N})$. This preserves coarse-grained routing structure without boundary artifacts when tokenization induces length mismatch.

Critically, practical transformations such as structured pruning, distillation, or architecture edits may change model structure (e.g., removing layers or attention heads), yielding fingerprints with different feature dimensions (e.g., $F \in \mathbb{R}^{M \times D}$ and $F' \in \mathbb{R}^{M \times D'}$ with $D \neq D'$ due to different L or H). This makes direct feature matching ill-posed. In contrast, CKA operates on probe-indexed Gram matrices in $\mathbb{R}^{M \times M}$ and therefore only requires the same probe set size M, enabling direct comparison without explicit layer/head alignment. Consequently, ATTNDIFF remains applicable across models with heterogeneous depths and head counts, while retaining robustness to re-parameterizations and other real-world modifications.

5 Experiment

We evaluate ATTNDIFF under realistic model transformations and compare it against representative non-invasive baselines: experimental setting in Sec. 5.1, fine-tuning in Sec. 5.2, merging in Sec. 5.3, pruning in Sec. 5.4, and ablations in Sec. 5.5. Figure 3 provides an aggregate view of ATTNDIFF's robustness across fine-tuning, merging, pruning, and additional evaluations including preference optimization (Appendix F.1) and distillation (Appendix F.5) compared to baseline methods.

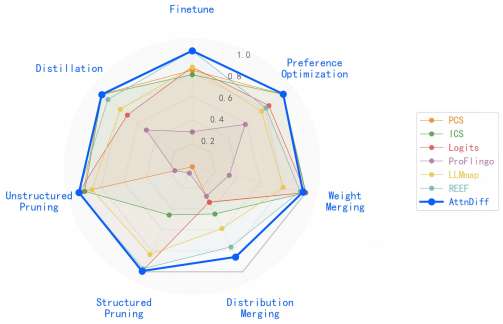


Figure 3: Radar comparison across deployment transformations. Larger radii indicate stronger robustness.

5.1 Experimental setting

Probe set. Unless otherwise specified, we construct a probe set with $M = 60$ prompts spanning six domains, with 10 prompts per domain: *Code*, *Math*, *Economics*, *Medicine*, *Daily QA*, and *Safe Alignment*. For each origin prompt p, we generate a minimally edited corrupted prompt $\tilde{p}$ via a single-word pivot and compute fingerprints on the resulting M probe pairs.

Model. We evaluate ATTNDIFF under realistic model transformation scenarios, covering diverse base architectures and a broad suite of derivative models. Our experiments cover models from the Llama-2/3 [46, 13], Qwen2.5 [38], Gemma [44], and Mistral [19] families, spanning parameter scales from 1B to 14B and including downstream adaptations and optimizations such as supervised fine-tuning, preference optimization, knowledge distillation, pruning, and model merging.

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Baselines. Following the taxonomy in the Introduction, we select representative *non-invasive* fingerprinting baselines that cover both black-box and white-box access assumptions. Specifically, we include PCS/ICS [66, 17] as *parameter-based* fingerprints, Logits [63] and REEF [67] as *representation-based* fingerprints, ProFlingo [21] as an *adversarial-example-based* fingerprint, and LLMMap [37] as a *semantic* fingerprint. For each method, we report the similarity score produced by its original formulation.

Our experiments aim to answer: **RQ1**: Is ATTNDIFF robust under downstream fine-tuning and preference optimization? **RQ2**: Is ATTNDIFF robust to models produced by diverse model merging strategies? **RQ3**: How robust is ATTNDIFF to diverse pruning/compression strategies?

5.2 Model fine-tuning

Table 1: SFT robustness results (similarity score) on Llama-2-7B-derived suspect models.

	Llama-2-Finance-7B (5M tokens)	Vicuna-1.5-7B (370M tokens)	WizardMath-7B (1.8B tokens)	ChineseLLaMA-2-7B (13B tokens)	CodeLLaMA-7B (500B tokens)	Llemma-7B (700B tokens)	Avg
PCS	0.9979	0.9985	0.9965	0.9390	0.5301	0.5052	0.8279
ICS	0.9952	0.9949	0.9985	0.7309	0.5112	0.5104	0.7902
Logits	0.9999	0.9999	0.9999	0.7033	0.7833	0.6367	0.8538
REEF	0.9950	0.9985	0.9979	0.9974	0.9947	0.9962	0.9966
ProFlingo	0.2400	0.5200	0.4200	0.2800	0.2000	0.1400	0.3000
LLMMap	0.8986	0.7294	0.7691	0.8720	0.9555	0.8998	0.8541
Ours	0.9989	0.9986	0.9985	0.9963	0.9890	0.9856	0.9945

Suspect model headers include the fine-tuning data scale. Cell shading (per column): =best, =second, =third.

Fine-tuning is a critical stage in the LLM lifecycle for adapting base models to downstream tasks or aligning them with human values [36, 39]. This process involves extensive parameter updates that can significantly shift the model's internal representations and output distributions, posing a severe test for fingerprint robustness [34].

For SFT, we use Llama-2-7B as the victim model and collect suspect models fine-tuned with different data scales and application domains, including Llama-2-finance-7b [33], Vicuna-1.5-7b [6], WizardMath-7b [30], Chinese-LLaMA-2-7b [7], CodeLLaMA-7b [41], and Llemma-7b [2].

Table 1 indicates that large-scale SFT can substantially alter model parameters/representations, causing parameter-based fingerprints (PCS/ICS) to drop sharply on heavily fine-tuned suspects. Logits also drops to 0.7833/0.6367 on CodeLLaMA-7B and Llemma-7B. In contrast, ATTNDIFF maintains uniformly high similarity (> 0.985) across all SFT suspects and is comparable to the SOTA baseline REEF.

5.3 Model merge

Model merging combines multiple pretrained models in weight space or at the distribution/behavior level, often integrating complementary capabilities without accessing training data or retraining [61, 1, 47, 48, 64, 18]. Because a merged suspect is derived from multiple victim models, its fingerprints can be mixed and harder to attribute to all sources.

From Table 2, ATTNDIFF consistently attains high similarity with all parent models in weight-space merges (≥ 0.95) and maintains substantial similarity (≈ 0.78–0.80) with heterogeneous distribution-level merges, whereas several baselines either fail to attribute all sources or exhibit noticeably weaker alignment on cross-architecture merges.

5.4 Model pruning

Model pruning compresses LLMs by removing redundant parameters to improve efficiency, often followed by retraining to recover capabilities. This poses a distinct challenge to model fingerprinting, as it fundamentally alters the model's weights and architecture. Recent studies have shown that pruning can weaken the verification effectiveness of multiple fingerprinting schemes [56, 52]. We

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Table 2: Model merge robustness results (similarity score) on open-source merged suspects.

	Weight Merging (Evollm-jp-7b)			Distribution Merging (Fusellm-7b)		
	Shisa-gamma-7b-v1	Wizardmath-7b-1.1	Abel-7b-002	Llama-2-7b	Openllama-2-7b	Mpt-7b
PCS	0.9992	0.9990	0.9989	0.9997	0.0194	0.0000
ICS	0.9992	0.9988	0.9988	0.9986	0.2478	0.1014
Logits	0.9933	0.9999	0.9999	0.9999	0.0100	0.0000
REEF	0.9635	0.9526	0.9374	0.9996	0.6713	0.6200
ProFlingo	0.3000	0.4000	0.2800	0.5400	0.1400	0.1600
LLMMap	0.7651	0.8343	0.8011	0.9511	0.5742	0.2413
Ours	0.9726	0.9561	0.9996	0.9962	0.7953	0.7851

Cell shading (per column): =*best*, =*second*, =*third*.

therefore evaluate robustness against a spectrum of pruning methodologies, including both structured and unstructured pruning on open-source checkpoints as well as additional suspects generated via the LLMPruner toolkit [31].

Table 3 shows that ATTNDIFF preserves high similarity across all structured and unstructured pruned suspects (no lower than 0.9879), even when aggressive structured pruning causes parameter-based fingerprints (PCS/ICS) to collapse and ProFlingo/LLMMap to degrade substantially. These results demonstrate that our differential attention fingerprint is robust to diverse pruning and compression strategies.

Table 3: Robustness results (similarity score) on pruned suspect models, comparing structured and unstructured pruning strategies.

	Structured Pruning						Unstructured Pruning		
	Sheared-llama-1.3b-pruned	Sheared-llama-1.3b	Sheared-llama-1.3b-sharegpt	Sheared-llama-2.7b-pruned	Sheared-llama-2.7b	Sheared-llama-2.7b-sharegpt	Sparse-llama-2-7b	Wanda-llama-2-7b	GBLM-llama-2-7b
PCS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.9560	0.9620	0.9616
ICS	0.4927	0.3512	0.3510	0.6055	0.4580	0.4548	0.9468	0.9468	0.9478
Logits	0.9967	0.9999	0.9999	0.9967	0.9999	0.9999	0.9999	0.9999	0.9999
REEF	0.9368	0.9676	0.9710	0.9278	0.9701	0.9991	0.9985	0.9986	0.9991
ProFlingo	0.0400	0.0200	0.0800	0.0200	0.1000	0.0800	0.1600	0.1200	0.1800
LLMMap	0.9088	0.9007	0.8152	0.9400	0.9236	0.7072	0.8459	0.9145	0.8956
Ours	0.9879	0.9938	0.9903	0.9929	0.9952	0.9936	0.9996	0.9927	0.9995

Cell shading (per column): =*best*, =*second*, =*third*.

5.5 Ablation study

We conduct ablation studies to validate key design choices of ATTNDIFF. Beyond the core experiments reported below, we perform extensive hyperparameter and design ablations in the appendix.

Differential mechanism (Table 11): comparing against a non-differential baseline that uses standard attention maps, the differential design is crucial under heavy domain adaptation—CodeLlama-7b and Llemma-7b drop to 0.51/0.29 without differencing but recover to 0.99/0.99 with ATTNDIFF ($\Delta > 0.47$). For unrelated models, differencing reduces spurious similarity from ~ 0.4 to < 0.23.

Probe configuration (Figure 5, Table 12): sparse probing ($M \leq 15$) yields high spurious similarity among unrelated models; $M = 60$ provides optimal discrimination. Domain-specific fine-tuning induces localized shifts in expert domains, but aggregated fingerprints remain stable for related suspects (> 0.98) and well separated from unrelated architectures (< 0.22).

Probe length (Appendix B.1, Table 9): medium-length prompts ($\ell \approx 30$ words) balance semantic context and attention stability.

Spectral rank (Appendix B.2, Table 10): $K = 3$ optimally trades off discriminability and compactness.

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Similarity metric (Appendix B.3, Table 14): CKA achieves superior discriminability over CCA/SVCCA with lower spurious similarity on unrelated models (0.1647 vs 0.3112/0.2419).

5.6 Discriminability analysis

While Sec. 5.2–5.4 demonstrate that ATTNDIFF maintains high similarity for related derivatives, practical fingerprinting also requires strong *discriminability*: the ability to reject unrelated models with high confidence. We therefore evaluate ATTNDIFF as a binary classifier that labels each suspect as either a derivative (positive) or unrelated (negative), and report both fixed-threshold and threshold-free discriminability metrics.

Table 4: Threshold-free discriminability (AUROC/AUPR, related=positive).

Method	AUROC	AUPR
PCS	0.9566	0.9367
ICS	0.9789	0.9795
Logits	0.9757	0.9746
REEF	0.9867	0.9877
ProFlingo	0.8374	0.8927
LLMMap	0.8925	0.9014
Ours	**0.9883**	**0.9881**

ATTNDIFF *attains highest AUROC/AUPR, indicating strong separability.*

Experimental setup. We construct two model pools: **(i) Related suspects** ($n = 36$, positive class): Llama-2-7B derivatives via fine-tuning, PPO/DPO, pruning/compression, and merging (Tables 1, 3, A1, A2, A3); **(ii) Unrelated suspects** ($n = 33$, negative class): distinct architectures/families from cross-family evaluation and ablations (Tables 11, A3, A4, A5).

For a threshold t, we predict *positive* (derivative) if $\mathrm{CKA}(F, F') \geq t$, otherwise *negative* (unrelated). We report two types of metrics: **(i) Fixed-threshold**—Precision (P), Recall (R), and F1-score (F1) at $t \in \{0.10, 0.25, 0.50, 0.90\}$; **(ii) Threshold-free**—AUROC and AUPR, treating related suspects as the positive class.

Table 5: Discriminability at fixed thresholds $t \in \{0.10, 0.25, 0.50, 0.90\}$: precision (P), recall (R), F1-score.

Method	$t=0.10$			$t=0.25$			$t=0.50$			$t=0.90$		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
PCS	0.6596	0.8611	0.7470	0.6739	0.8611	0.7561	0.8108	0.8333	0.8219	0.8788	**0.8056**	0.8406
ICS	0.5714	**1.0000**	0.7273	0.8649	0.8889	0.8767	0.9394	0.8611	0.8986	0.9630	0.7222	0.8254
Logits	0.5556	0.9722	0.7071	0.7778	0.9722	0.8642	0.8947	**0.9444**	**0.9189**	0.9655	0.7778	0.8615
REEF	0.6545	**1.0000**	**0.7912**	0.8611	0.8611	0.8611	**1.0000**	0.7778	0.8750	**1.0000**	0.6389	0.7797
ProFlingo	**0.7778**	0.7778	0.7778	**0.9524**	0.5556	0.7018	**1.0000**	0.2222	0.3636	**1.0000**	0.1389	0.2439
LLMMap	0.5217	**1.0000**	0.6857	0.6481	0.9722	0.7778	0.7143	0.8333	0.7692	0.8571	0.3333	0.4800
Ours	0.5455	**1.0000**	0.7059	0.8182	**1.0000**	**0.9000**	0.8500	**0.9444**	**0.9189**	**1.0000**	0.7778	**0.8750**

ATTNDIFF *achieves best F1 at* $t = 0.25$ *(0.9000).*

Results and analysis. ATTNDIFF achieves the highest F1 at $t = 0.25$ (0.9000) and $t = 0.90$ (0.8750). At $t = 0.25$, it is the only method with perfect recall (R=1.0000) while maintaining strong precision (P=0.8182), effectively balancing zero missed derivatives with controlled false positives. At $t = 0.90$, it achieves perfect precision (P=1.0000, FP=0) with the highest F1, demonstrating robust separability across practical operating points.

Table 4 further confirms this via threshold-free metrics: ATTNDIFF attains the highest AUROC (0.9883) and AUPR (0.9881), outperforming all baselines. Notably, REEF is competitive (0.9867/0.9877), while ProFlingo and LLMMap drop to ≈ 0.83–0.90, reflecting their sensitivity to surface-level transformations. Together, these results demonstrate that ATTNDIFF provides a practical and discriminative tool for binary provenance decisions across realistic operating points.

6 Discussion

ATTNDIFF supports post-hoc provenance verification under common laundering pipelines, including fine-tuning, preference optimization, pruning/compression, and merging, while maintaining a clear separation margin from unrelated architectures.

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Contributions. We propose a white-box, post-hoc fingerprinting method that captures model-specific information-routing behavior via *differential attention* under controlled semantic conflicts. We represent each model with compact spectral descriptors and compare fingerprints using centered linear CKA, enabling architecture-agnostic similarity that remains stable for co-originated derivatives and well separated from unrelated models under common laundering operations.

Probe refresh. Beyond average-case robustness, ATTNDIFF is compatible with operational probe refresh: a defender can update or expand the probe set to reduce attack transfer without changing the reference model. This is important in post-hoc auditing settings where a suspect may have been adapted after release and where the defender can recompute fingerprints on demand.

Computational cost. ATTNDIFF requires no training or gradient-based optimization. With M probe pairs, fingerprint extraction requires $2M$ forward passes per model (origin + corrupted) while recording attention; post-processing (pooling when needed, differencing, top-K SVD, and CKA) is lightweight. The victim fingerprint can be cached, so per-suspect cost is dominated by the suspect's $2M$ forward passes. On a single NVIDIA RTX 5090 GPU, ATTNDIFF fingerprinting with $M = 60$ probe pairs takes approximately 5 minutes for a 7B-parameter model. Compared with baselines, ATTNDIFF incurs modest inference cost without heavyweight operations such as tensor construction and alignment heuristics (PCS/ICS), trigger optimization (ProFlingo), or inference model training (LLMMap). A detailed computational comparison is provided in Appendix D.3.

7 Conclusion

We present ATTNDIFF, a white-box post-hoc fingerprinting method that captures model-specific information-routing behavior via *differential attention* under controlled semantic conflicts. Using compact spectral descriptors and centered linear CKA, it yields model-agnostic fingerprints that remain stable for co-originated derivatives and well separated from unrelated architectures across fine-tuning (including PPO/DPO), pruning/compression, and model merging. With a small probe set and lightweight computation, ATTNDIFF provides a practical building block for provenance verification and accountability in the open-weight LLM ecosystem. Nevertheless, the adoption of such provenance tools also entails societal risks, such as the potential for unwarranted copyright disputes (see Appendix G for a detailed discussion).

Limitations and future work

White-box access. ATTNDIFF currently assumes a **white-box** setting: extracting fingerprints requires access to internal attention activations or equivalent hidden-state signals. Consequently, the method does not directly apply to strictly black-box APIs where only model outputs are observable.

Theoretical modeling of laundering effects. While the centered linear CKA comparison provides useful invariance to feature rotations, permutations, and isotropic scaling, we still lack a complete mathematical model that propagates common laundering operations (e.g., fine-tuning, pruning, merging, and distillation) from parameter or architecture transformations to perturbations in the extracted ATTNDIFF fingerprints and the resulting similarity scores. Establishing such an end-to-end modeling framework remains an important direction for future work.

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References

[1] Takuya Akiba, Makoto Shing, Yujin Tang, Qi Sun, and David Ha. Evolutionary optimization of model merging recipes. *arXiv preprint arXiv:2403.13187*, 2024.

[2] Zhangir Azerbayev, Hailey Schoelkopf, Keiran Paster, Marco Dos Santos, Stephen McAleer, Albert Q. Jiang, Jia Deng, Stella Biderman, and Sean Welleck. Llemma: An open language model for mathematics, 2023.

[3] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, Binyuan Hui, Luo Ji, Mei Li, Junyang Lin, Runji Lin, Dayiheng Liu, Gao Liu, Chengqiang Lu, Keming Lu, Jianxin Ma, Rui Men, Xingzhang Ren, Xuancheng Ren, Chuanqi Tan, Sinan Tan, Jianhong Tu, Peng Wang, Shijie Wang, Wei Wang, Shengguang Wu, Benfeng Xu, Jin Xu, An Yang, Hao Yang, Jian Yang, Shusheng Yang, Yang Yao, Bowen Yu, Hongyi Yuan, Zheng Yuan, Jianwei Zhang, Xingxuan Zhang, Yichang Zhang, Zhenru Zhang, Chang Zhou, Jingren Zhou, Xiaohuan Zhou, and Tianhang Zhu. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023. URL `https://arxiv.org/abs/2309.16609`.

[4] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[5] Jiacheng Cai, Jiahao Yu, Yangguang Shao, Yuhang Wu, and Xinyu Xing. UTF: Under-trained tokens as fingerprints—a novel approach to LLM identification. In *Proceedings of the First Workshop on LLM Security (LLMSEC)*, pages 1–6, 2025.

[6] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, et al. Vicuna: An open-source chatbot impressing GPT-4 with 90%* ChatGPT quality, March 2023. URL `https://lmsys.org/blog/2023-03-30-vicuna/`.

[7] Yiming Cui, Ziqing Yang, and Xin Yao. Efficient and effective text encoding for chinese llama and alpaca, 2023. URL `https://github.com/ymcui/Chinese-LLaMA-Alpaca`.

[8] MohammadReza Davari and Eugene Belilovsky. Model breadcrumbs: Scaling multi-task model merging with sparse masks. In *Computer Vision – ECCV 2024*, 2024.

[9] Pala Tej Deep, Rishabh Bhardwaj, and Soujanya Poria. Della-merging: Reducing interference in model merging through magnitude-based sampling. *arXiv preprint arXiv:2406.11617*, 2024.

[10] Pierre Fernandez, Guillaume Couairon, Teddy Furon, and Matthijs Douze. Functional invariants to watermark large transformers. In *ICASSP 2024*, 2023.

[11] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.

[12] Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*, 2022.

[13] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[14] Martin Gubri, Dennis Thomas Ulmer, Hwaran Lee, Sangdoo Yun, and Seong Joon Oh. TRAP: Targeted random adversarial prompt honeypot for black-box identification. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11496–11517, 2024.

[15] Qingxiao Guo, Xinjie Zhu, Yilong Ma, Hui Jin, Yunhao Wang, Weifeng Zhang, and Xiaobing Guo. Invariant-based robust weights watermark for large language models. *arXiv preprint arXiv:2507.08288*, 2025.

[16] Harold Hotelling. Relations between two sets of variates. *Biometrika*, 28(3/4):321–377, 1936.

Confidential reviewer copy. This manuscript is submitted to the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

[17] Do hyeon Yoon, Minsoo Chun, Thomas Allen, Hans Müller, Min Wang, and Rajesh Sharma. Intrinsic fingerprint of LLMs: Continue training is NOT all you need to steal a model! *arXiv preprint arXiv:2507.03014*, 2026. URL https://arxiv.org/abs/2507.03014.

[18] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Wadden, Hannaneh Hajishirzi, Dani Yogatama, and Luke Zettlemoyer. Editing models with task arithmetic. In *The Eleventh International Conference on Learning Representations*, 2023.

[19] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. URL https://arxiv.org/abs/2310.06825.

[20] Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, Lélio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Théophile Gervet, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mixtral of experts, 2024. URL https://arxiv.org/abs/2401.04088.

[21] Heng Jin, Chaoyu Zhang, Shanghao Shi, Wenjing Lou, and Y Thomas Hou. Proflingo: A fingerprinting-based intellectual property protection scheme for large language models. In *2024 IEEE Conference on Communications and Network Security (CNS)*, pages 1–9. IEEE, 2024.

[22] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529. PMLR, 2019.

[23] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.

[24] Quentin Lhoest, Clément Delangue, Patrick von Platen, Thomas Wolf, et al. Datasets: A community library for natural language processing. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2021.

[25] Peixuan Li, Pengzhou Cheng, Fangqi Li, Wei Du, Haodong Zhao, and Gongshen Liu. Plmmark: A secure and robust black-box watermarking framework for pre-trained language models. In *Proceedings of the AAAI Conference on Artificial Intelligence 2023*, pages 14991–14999, 2023.

[26] Shen Li, Liuyi Yao, Jinyang Gao, Lan Zhang, and Yaliang Li. Double-I watermark: Protecting model copyright for LLM fine-tuning. *arXiv preprint arXiv:2402.14883*, 2024. URL https://arxiv.org/abs/2402.14883.

[27] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *arXiv preprint arXiv:2109.07958*, 2021.

[28] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.

[29] Xiaokun Luan, Zeming Wei, Yihao Zhang, and Meng Sun. Robust and efficient watermarking of large language models using error correction codes. *Proceedings on Privacy Enhancing Technologies*, 2025.

[30] Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Jianguang Lou, Chongyang Tao, Xiubo Geng, Qingwei Lin, Shifeng Chen, and Dongmei Zhang. Wizardmath: Empowering mathematical reasoning for large language models via reinforced evol-instruct, 2023.

Confidential reviewer copy. This manuscript is submitted to the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

[31] Xinyin Ma, Gongfan Fang, and Xinchao Wang. LLM-pruner: On the structural pruning of large language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 21702–21720. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/44956951349095f74492a5471128a7e0-Paper-Conference.pdf.

[32] N Gregory Mankiw. *Principles of economics*. Cengage Learning, 2020.

[33] Meta AI. Llama 2 community license agreement, 2023. URL https://ai.meta.com/llama/license/. Accessed: 2024-08-28.

[34] Anshul Nasery, Edoardo Contente, Alkin Kaz, Pramod Viswanath, and Sewoong Oh. Are robust LLM fingerprints adversarially robust? *arXiv preprint arXiv:2509.26598*, 2025.

[35] Anshul Nasery, Jonathan Hayase, Creston Brooks, Peiyao Sheng, Himanshu Tyagi, Pramod Viswanath, and Sewoong Oh. Scalable fingerprinting of large language models. *arXiv preprint arXiv:2502.07760*, 2025.

[36] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.

[37] Dario Pasquini, Evgenios M Kornaropoulos, and Giuseppe Ateniese. LLMmap: Fingerprinting for Large Language Models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 299–318, 2025.

[38] Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025. URL https://arxiv.org/abs/2412.15115.

[39] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36:53728–53741, 2023.

[40] Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[41] Baptiste Roziere, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, et al. Code llama: Open foundation models for code, 2023.

[42] Mark Russinovich and Ahmed Salem. Hey, that's my model! introducing chain & hash, an LLM fingerprinting technique. *arXiv preprint arXiv:2407.10887*, 2024.

[43] Ling Tang, Yuefeng Chen, Hui Xue, and Quanshi Zhang. Towards the resistance of neural network watermarking to fine-tuning. *arXiv preprint arXiv:2505.01007*, 2025. URL https://arxiv.org/abs/2505.01007.

[44] Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, et al. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.

[45] Yao Tong, Haonan Wang, Siquan Li, Kenji Kawaguchi, and Tianyang Hu. Seedprints: Fingerprints can even tell which seed your large language model was trained from. *arXiv preprint arXiv:2509.26404*, 2025.

Confidential reviewer copy. This manuscript is submitted to the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

[46] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

[47] Fanqi Wan, Xinting Huang, Deng Cai, Xiaojun Quan, Wei Bi, and Shuming Shi. Knowledge Fusion of Large Language Models, 2024. URL https://arxiv.org/abs/2401.10491.

[48] Fanqi Wan, Ziyi Yang, Longguang Zhong, Canbin Huang, Guosheng Liang, and Xiaojun Quan. Fusechat: Knowledge fusion of chat models. *arXiv preprint arXiv:2408.07990*, 2024.

[49] Zhebo Wang, Zhenhua Xu, Maike Li, Wenpeng Xing, Chunqiang Hu, Chen Zhi, and Meng Han. SRAF: Stealthy and robust adversarial fingerprint for copyright verification of large language models. *arXiv preprint arXiv:2505.06304*, 2026. URL https://arxiv.org/abs/2505.06304.

[50] Zhaomin Wu, Haodong Zhao, Ziyang Wang, Jizhou Guo, Qian Wang, and Bingsheng He. LLM DNA: Tracing model evolution via functional representations. *arXiv preprint arXiv:2509.24496*, 2025.

[51] Jiashu Xu, Fei Wang, Mingyu Ma, Pang Wei Koh, Chaowei Xiao, and Muhao Chen. Instructional fingerprinting of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3277–3306, 2024.

[52] Zhenhua Xu, Meng Han, and Wenpeng Xing. EverTracer: Hunting stolen large language models via stealthy and robust probabilistic fingerprint. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7019–7042, 2025.

[53] Zhenhua Xu, Meng Han, Xubin Yue, and Wenpeng Xing. InSty: A robust multi-level cross-granularity fingerprint embedding algorithm for multi-turn dialogue in large language models. *SCIENTIA SINICA Informationis*, 55(8):1906, 2025. ISSN 1674-7267. doi: 10.1360/SSI-2025-0022.

[54] Zhenhua Xu, Qichen Liu, Zhebo Wang, Wenpeng Xing, Dezhang Kong, Mohan Li, and Meng Han. Fingerprint vector: Enabling scalable and efficient model fingerprint transfer via vector addition. *arXiv preprint arXiv:2409.08846*, 2025. URL https://arxiv.org/abs/2409.08846.

[55] Zhenhua Xu, Zhaokun Yan, Binhan Xu, Xin Tong, Haitao Xu, Yourong Chen, and Meng Han. Unlocking the Effectiveness of LoRA-FP for Seamless Transfer Implantation of Fingerprints in Downstream Models. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4302–4312, Suzhou, China, 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.230.

[56] Zhenhua Xu, Xixiang Zhao, Xubin Yue, Shengwei Tian, Changting Lin, and Meng Han. CTCC: A robust and stealthy fingerprinting framework for large language models via cross-turn contextual correlation backdoor. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 6978–7000, 2025.

[57] Zhenhua Xu, Xubin Yue, Zhebo Wang, Haobo Zhang, Qichen Liu, Xixiang Zhao, Jingxuan Zhang, Wenjun Zeng, Wengpeng Xing, Dezhang Kong, Changting Lin, and Meng Han. Copyright protection for large language models: A survey of methods, challenges, and trends. *arXiv preprint arXiv:2508.11548*, 2026. URL https://arxiv.org/abs/2508.11548.

[58] Zhenhua Xu, Haobo Zhang, Zhebo Wang, Qichen Liu, Haitao Xu, Wenpeng Xing, and Meng Han. Forgetmark: Stealthy fingerprint embedding via targeted unlearning in language models. *arXiv preprint arXiv:2601.08189*, 2026. URL https://arxiv.org/abs/2601.08189.

[59] Zhenhua Xu, Yiran Zhao, Mengting Zhong, Dezhang Kong, Changting Lin, Tong Qiao, and Meng Han. DNF: Dual-layer nested fingerprinting for large language model intellectual property protection. *arXiv preprint arXiv:2601.08223*, 2026. URL https://arxiv.org/abs/2601.08223.

Confidential reviewer copy. This manuscript is submitted to the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Unauthorized sharing, redistribution, or disclosure is strictly prohibited.

[60] Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. Ties-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, 2023.

[61] Enneng Yang, Li Shen, Guibing Guo, Xingwei Wang, Xiaochun Cao, Jie Zhang, and Dacheng Tao. Model merging in LLMs, MLLMs, and beyond: Methods, theories, applications and opportunities. *arXiv preprint arXiv:2408.07666*, 2024.

[62] Kang Yang, Run Wang, and Lina Wang. Metafinger: Fingerprinting the deep neural networks with meta-training. In *IJCAI*, pages 776–782, 2022.

[63] Zhiguang Yang and Hanzhou Wu. A Fingerprint for Large Language Models. *arXiv preprint arXiv:2407.01235*, 2024. URL `https://arxiv.org/abs/2407.01235`.

[64] Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 57911–57932. PMLR, 2024.

[65] Xubin Yue, Zhenhua Xu, Wenpeng Xing, Jiahui Yu, Mohan Li, and Meng Han. PREE: Towards Harmless and Adaptive Fingerprint Editing in Large Language Models via Knowledge Prefix Enhancement. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 3794–3804, Suzhou, China, 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.204.

[66] Boyi Zeng, Lizheng Wang, Yuncong Hu, Yi Xu, Chenghu Zhou, Xinbing Wang, Yu Yu, and Zhouhan Lin. HuRef: Human-readable fingerprint for large language models. *Advances in Neural Information Processing Systems*, 37:126332–126362, 2024.

[67] Jie Zhang, Dongrui Liu, Chen Qian, Linfeng Zhang, Yong Liu, Yu Qiao, and Jing Shao. REEF: Representation encoding fingerprints for large language models. In *The Thirteenth International Conference on Learning Representations*.

[68] Jingxuan Zhang, Zhenhua Xu, Rui Hu, Wenpeng Xing, Xuhong Zhang, and Meng Han. MEraser: An effective fingerprint erasure approach for large language models. *arXiv preprint arXiv:2506.12551*, 2025. URL `https://arxiv.org/abs/2506.12551`.

[69] Ruisi Zhang and Farinaz Koushanfar. Emmark: Robust watermarks for IP protection of embedded quantized large language models. In *Proceedings of the 61st ACM/IEEE Design Automation Conference*, pages 1–6, 2024.

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A Theoretical guarantees of ATTNDIFF

We provide a theoretical analysis of ATTNDIFF centered on the centered linear CKA similarity. Our goal is to justify: (i) why centered linear CKA is suitable for comparing ATTNDIFF fingerprints even when the feature dimensions differ ($D \neq D'$); and (ii) why common post-training transformations often preserve high similarity for related models.

Roadmap.

- **CKA preliminaries (Appendix. A.1).** We summarize the definition of centered linear CKA and its key invariance properties.
- **Compatibility of ATTNDIFF with CKA (Appendix. A.2).** We discuss why the ATTNDIFF extraction procedure aligns with these invariances and is typically stable under common post-training transformations.
- **Perturbation stability (Appendix. A.3).** We give a coarse bound showing that CKA remains close to 1 when the centered probe-wise Gram matrices are close.

Together, these results clarify why CKA is suitable for comparing ATTNDIFF fingerprints across heterogeneous feature dimensions.

A.1 Centered linear CKA: preliminaries

Objects. Let $F \in \mathbb{R}^{M\times D}$ and $F' \in \mathbb{R}^{M\times D'}$ be fingerprint matrices computed on the same M probes. We compare fingerprints through Gram matrices $K = FF^\top$ and $K' = F'F'^\top \in \mathbb{R}^{M\times M}$, which only require matching M while allowing $D \neq D'$.

Definition (Centered linear CKA). Let $H = I - \frac{1}{M}\mathbf{1}\mathbf{1}^\top$ be the centering matrix and define the centered Gram matrices $\bar{K} = HKH$ and $\bar{K}' = HK'H$. The (biased) HSIC and centered linear CKA are

$$\begin{aligned}\mathrm{HSIC}(K,K') &:= \tfrac{1}{(M-1)^2}\,\mathrm{tr}(\bar{K}\,\bar{K}'),\\ \mathrm{CKA}(F,F') &:= \frac{\mathrm{HSIC}(K,K')}{\sqrt{\mathrm{HSIC}(K,K)\,\mathrm{HSIC}(K',K')}}\\ &= \frac{\langle \bar{K},\bar{K}'\rangle_F}{\|\bar{K}\|_F\,\|\bar{K}'\|_F},\end{aligned}$$

where $\langle A,B\rangle_F := \mathrm{tr}(A^\top B)$ denotes the Frobenius inner product.

Proposition 1 (Invariance of centered linear CKA). For any permutation matrix S (simultaneous probe/row permutation), any permutation matrices P, P' (column permutations), any orthogonal matrices Q, Q' (feature rotations), and any scalars $\alpha, \beta > 0$, centered linear CKA satisfies

$$\begin{aligned}\mathrm{CKA}(F,F') &= \mathrm{CKA}(FP, F'P')\\ &= \mathrm{CKA}(FQ, F'Q')\\ &= \mathrm{CKA}(\alpha F, \beta F')\\ &= \mathrm{CKA}(SF, SF').\end{aligned}$$

Proof. Centered linear CKA depends on F only through the centered Gram matrix $\bar{K} = HFF^\top H$ (and likewise F' through $\bar{K}' = HF'F'^\top H$). **(i) Probe permutations.** Let S be any $M \times M$ permutation matrix. Since $S\mathbf{1} = \mathbf{1}$, we have $SHS^\top = H$. Moreover, the Gram matrix becomes $K_S = (SF)(SF)^\top = SKS^\top$, so the centered Gram matrix satisfies

$$\bar{K}_S = HK_SH = HSKS^\top H = (SHS^\top)(SKS^\top)(SHS^\top) = S(HKH)S^\top = S\bar{K}S^\top,$$

and similarly $\bar{K}'_S = S\bar{K}'S^\top$. Because Frobenius inner products and norms are invariant under simultaneous conjugation by a permutation matrix, we get

$$\frac{\langle \bar{K}_S, \bar{K}'_S\rangle_F}{\|\bar{K}_S\|_F\,\|\bar{K}'_S\|_F} = \frac{\langle \bar{K}, \bar{K}'\rangle_F}{\|\bar{K}\|_F\,\|\bar{K}'\|_F},$$

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which implies $\mathrm{CKA}(F, F') = \mathrm{CKA}(SF, SF')$. **(ii) Orthogonal transforms.** Let R be any orthogonal matrix (so $RR^\top = I$; this includes permutations and rotations). Then $(FR)(FR)^\top = FRR^\top F^\top = FF^\top$, so K (and hence $\bar{K} = HKH$) is unchanged. Applying the same argument to F' yields $\mathrm{CKA}(F, F') = \mathrm{CKA}(FR, F'R')$, which covers the equalities for P, P' and Q, Q'. **(iii) Isotropic scaling.** For $\alpha > 0$, $(\alpha F)(\alpha F)^\top = \alpha^2 FF^\top$, hence $H(\alpha^2 K)H = \alpha^2 HKH = \alpha^2 \bar{K}$ by linearity of centering; similarly $\bar{K}'$ scales to $\beta^2 \bar{K}'$. Plugging into the normalized inner-product form of CKA gives

$$\mathrm{CKA}(\alpha F, \beta F') = \frac{\alpha^2 \beta^2 \langle \bar{K}, \bar{K}' \rangle_F}{\alpha^2 \beta^2 \|\bar{K}\|_F \, \|\bar{K}'\|_F} = \mathrm{CKA}(F, F').$$

Remark. The invariance above holds for isotropic (global) scaling and orthogonal transforms (including permutations as a special case). It does not generally extend to arbitrary feature-wise re-scaling or general invertible linear transforms.

A.2 Robustness of AttnDiff fingerprint extraction

AttnDiff robustness stems from three design principles: (i) differential attention $\Delta A = \tilde{A} - A$ suppresses prompt-specific baselines, (ii) SVD-based $\texttt{TopK}(\sigma)$ descriptors capture dominant re-routing structure, and (iii) centered linear CKA measures similarity in probe-indexed Gram space. These components collectively yield invariance properties that enable stable provenance verification.

Differential cancellation. Attention maps decompose into prompt-dependent baselines (token positions, syntax, domain priors) plus conflict-induced re-routing. Computing $\Delta A = \tilde{A} - A$ cancels shared baselines and isolates the re-routing signal. Under adaptation (e.g., domain-specific fine-tuning), baselines shift but the differential re-routing pattern—governed by intrinsic information-flow architecture—remains stable, reducing sensitivity to domain drift.

Spectral invariances. Singular values of ΔA are invariant to orthogonal token-space transformations and capture dominant energy distribution. Under transformations preserving the routing subspace (near-orthogonal re-parameterizations from weight updates, global scaling from normalization), fingerprints F undergo rotations/permutations/isotropic scaling. Proposition 1 shows such changes leave centered linear CKA invariant.

Visualization. For heatmaps, we compute rank-K spectral descriptors (top-K singular values) at each layer l and head h, yielding $T \in \mathbb{R}^{L \times H \times K}$ ($K = 3$). We aggregate via $E_{l,h} = \|T_{l,h,:}\|_2$ to produce $L \times H$ energy maps. Figure 4 shows stability across transformations.

Perturbation propagation. Since AttnDiff compares via Gram matrices $K = FF^\top$ and $K' = F'F'^\top$, fingerprint perturbations control Gram perturbations:

$$\begin{aligned}\|K - K'\|_F = \|FF^\top - F'F'^\top\|_F &\le \|(F - F')F^\top\|_F + \|F'(F - F')^\top\|_F \\ &\le \|F - F'\|_F \, \|F\|_F + \|F'\|_F \, \|F - F'\|_F = \|F - F'\|_F \, (\|F\|_F + \|F'\|_F).\end{aligned}$$

Centering is non-expansive: with $\bar{K} = HKH$ and $\bar{K}' = HK'H$,

$$\|\bar{K} - \bar{K}'\|_F = \|H(K - K')H\|_F \le \|K - K'\|_F,$$

since H is an orthogonal projection ($\|H\|_2 = 1$). Thus centering preserves stability. The next subsection bounds CKA degradation under Gram perturbations.

A.3 Stability of CKA under Gram perturbations

Proposition 1 characterizes exact invariances (e.g., global scaling and orthogonal feature re-parameterizations). In practice, post-training transformations may still induce residual changes in the probe-wise geometry encoded by centered Gram matrices. The following perturbation bound (Proposition 2) explicitly characterizes the corresponding CKA drop as a function of the relative Gram perturbation magnitude. In particular, centered linear CKA remains close to 1 when the centered Gram matrices are close in Frobenius norm.

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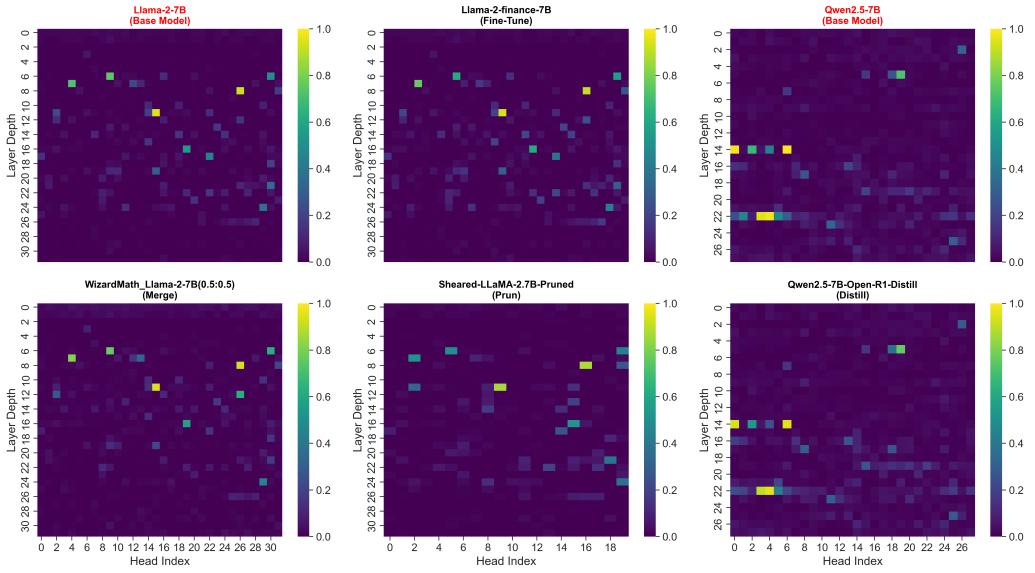


Figure 4: Visualization of single-sample layer-head fingerprint heatmaps ($L \times H$) under fine-tuning, pruning, merging, and distillation. Differential attention patterns remain visually consistent across laundering operations.

To connect with the experimental protocol, we measure the relative centered Gram perturbation by

$$\varepsilon := \|\bar{K} - \bar{K}'\|_F / \|\bar{K}\|_F,$$

where $\bar{K} = HKH$ and $\bar{K}' = HK'H$ are the centered probe-wise Gram matrices.

Proposition 2 (A coarse CKA perturbation bound). Assume $\bar{K} \neq 0$ and $\bar{K}' \neq 0$. If for some $\varepsilon \in (0, 1)$,

$$\|\bar{K} - \bar{K}'\|_F \leq \varepsilon \, \|\bar{K}\|_F,$$

then

$$1 - \mathrm{CKA}(F, F') \leq 2\varepsilon^2.$$

Proof. Let $a = \mathrm{vec}(\bar{K}) \in \mathbb{R}^{M^2}$ and $b = \mathrm{vec}(\bar{K}') \in \mathbb{R}^{M^2}$. Define unit vectors $u = a/\|a\|_2$ and $v = b/\|b\|_2$, so that $\mathrm{CKA}(F, F') = \langle u, v \rangle$. The assumption implies

$$\|a - b\|_2 = \|\bar{K} - \bar{K}'\|_F \leq \varepsilon \|\bar{K}\|_F = \varepsilon \|a\|_2.$$

Since $\varepsilon < 1$ and $a \neq 0$, this also implies $b \neq 0$. We bound the distance between normalized vectors as

$$\begin{aligned} \|u - v\|_2 &= \left\| \frac{a}{\|a\|_2} - \frac{b}{\|b\|_2} \right\|_2 \leq \frac{\|a - b\|_2}{\|a\|_2} + \|b\|_2 \left| \frac{1}{\|a\|_2} - \frac{1}{\|b\|_2} \right| \\ &= \frac{\|a - b\|_2}{\|a\|_2} + \frac{\big|\|b\|_2 - \|a\|_2\big|}{\|a\|_2} \leq \frac{2\|a - b\|_2}{\|a\|_2} \leq 2\varepsilon. \end{aligned}$$

Finally, since $\|u\|_2 = \|v\|_2 = 1$, we have

$$\begin{aligned} \|u - v\|_2^2 &= 2 - 2\langle u, v \rangle, \\ 1 - \mathrm{CKA}(F, F') &= 1 - \langle u, v \rangle = \tfrac{1}{2}\|u - v\|_2^2 \leq 2\varepsilon^2. \end{aligned}$$

Quantitative illustration. We report representative ε values computed from the centered probe-wise Gram matrices in Table 6. Related derivatives typically yield small ε, whereas unrelated model families yield ε close to 1, matching the separation behavior predicted by Proposition 2.

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Summary. Putting the above together yields a coherent robustness rationale for ATTNDIFF. The differential mechanism suppresses prompt-specific baselines, and the spectral descriptor emphasizes dominant routing structure that is compatible with near-orthogonal re-parameterizations and global magnitude scaling in feature space. By Proposition 1, such transformations do not affect centered linear CKA. For the remaining residual drift, Proposition 2 shows that when the centered probe-wise Gram matrices are close (small ε), $\mathrm{CKA}(F, F')$ stays close to 1; Table 6 illustrates this regime for related derivatives. Conversely, unrelated model families yield ε near 1 and thus low CKA, providing a strong margin that mitigates false positives.

This theoretical perspective motivates the implementation choices detailed next in Appendix. B.

Table 6: Representative ε values under different model transformations, where $\varepsilon := \|\bar{K} - \bar{K}'\|_F / \|\bar{K}\|_F$ is computed from centered probe-wise Gram matrices. Related models exhibit small ε while unrelated models yield $\varepsilon \approx 1$, which explains the large margin in CKA and reduces false positives in provenance decisions.

Transformation Type	Model Example	CKA Score	$1-$CKA	ε	$2\varepsilon^2$ (bound)	Bound Validity
Fine-tuning	Llama-2-7B → WizardMath-7B	0.9985	0.0015	0.0777	0.0121	Valid
	Llama-2-7B → Llemma-7B	0.9856	0.0144	0.1771	0.0627	Valid
	Llama-2-7B → CodeLLaMA-7B	0.9890	0.0110	0.1625	0.0528	Valid
Model Merging	Llama-2-7B → Breadcrumbs-Llama-2-7B	0.9992	0.0008	0.0564	0.0064	Valid
	Llama-2-7B → Breadcrumbs+Ties-Llama-2-7B	0.9992	0.0008	0.0564	0.0064	Valid
	Llama-2-7B → Della-Llama-2-7B	0.9986	0.0014	0.0774	0.0120	Valid
	Llama-2-7B → Task-Llama-2-7B	0.9996	0.0004	0.0406	0.0033	Valid
Pruning	Llama-2-7B → Sheared-llama-1.3b-pruned	0.9879	0.0121	0.0972	0.0189	Valid
	Llama-2-7B → Sheared-llama-1.3b	0.9938	0.0062	0.0866	0.0150	Valid
	Llama-2-7B → Sheared-llama-2.7b-pruned	0.9929	0.0071	0.0758	0.0115	Valid
	Llama-2-7B → Sheared-llama-2.7b	0.9952	0.0048	0.0697	0.0097	Valid
Distillation	Qwen2.5-7B → Qwen2.5-7B-Open-R1-Distill	0.9873	0.0127	0.1138	0.0259	Valid
	Llama-2-7B → logit-watermark-distill	0.9998	0.0002	0.0395	0.0031	Valid
Unrelated	Llama-2-7B → gemma-2-2b	0.2154	0.7846	0.9984	1.9936	Valid
	Llama-2-7B → Qwen2.5-14B	0.1052	0.8948	0.9757	1.9040	Valid
	Llama-2-7B → Llama3-8B	0.2299	0.7701	0.9856	1.9428	Valid

B Implementation Details

This section provides implementation details necessary to reproduce ATTNDIFF, summarizing the end-to-end workflow in Alg. 1 with default hyperparameters in Table 7.

We detail three pipeline stages: probe construction (Appendix. B.1), fingerprint extraction (Appendix. B.2), and similarity measurement (Appendix. B.3).

B.1 Probe Construction Pipeline

We construct probes through three stages: source selection, filtering, and pivot injection.

Source Selection. We curate prompts from domain-specific sources to ensure comprehensive coverage across technical and general knowledge domains. For Math, Economics, and Medicine domains, sentences are extracted from standard textbooks, Wikipedia "Fact" sections [24], and authoritative reviews such as CDC guidelines and *Principles of Economics* [32]. Code-related probes are sourced from Python official documentation and high-voted Stack Overflow answers, with emphasis on boolean logic and control flow constructs. Daily QA prompts consist of common sense questions selected from Natural Questions [23] and TruthfulQA [27]. Finally, Safe Alignment probes are derived from Constitutional AI principles [4] and Anthropic's red-teaming datasets [12] to capture policy-gated behaviors and refusal patterns.

Filtering. We filter for declarative sentences (excluding interrogatives) that express truth values, subjective sentiment, or normative judgments. To reduce length-induced variation before pooling, we constrain the *word count* to be within ± 5 of a target length ℓ (in words). This step serves as a relaxed pre-alignment: it does not enforce exact token-length equality ($N = \tilde{N}$), but instead aims to keep the tokenizer-induced lengths N and $\tilde{N}$ (token counts under the model tokenizer) in a similar range so that the subsequent pooling operates under comparable resolutions. Concretely, we implement pivot

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Table 7: Default hyperparameters for AttnDiff.

Symbol	Meaning	Setting
M	# probe pairs	60
L	# transformer layers	all layers (model-dependent)
H	# attention heads per layer	all heads (model-dependent)
K	spectral rank (top singular values)	3 (Table 10)
$N, \tilde{N}$	token lengths of $p, \tilde{p}$	variable; controlled but not forced equal
N^*	pooling resolution	$\min(N, \tilde{N})$

Algorithm 1 ATTNDIFF Workflow

PHASE 1: Probe Generation
In: probe prompts, pivot rule **Out:** origin/corrupted pairs
for $i \leftarrow 1$ to M **do**
  Generate $\tilde{p}_i$ from p_i using pivot rule; apply length control
end for
PHASE 2: Fingerprint Extraction
In: models f, f', probe pairs, layers L, heads H, rank K **Out:** fingerprints
for $g \in \{f, f'\}$ **do**
  for $i \leftarrow 1$ to M **do**
    for $l \leftarrow 1$ to L **do**
      for $h \leftarrow 1$ to H **do**
        Compute differential attention between p_i and $\tilde{p}_i$
        Extract rank-K spectral descriptor
      end for
    end for
    Concatenate descriptors across layers/heads
  end for
  Stack descriptors to form fingerprint
end for
PHASE 3: Similarity Computation
In: fingerprints of f, f' **Out:** CKA similarity score s
Compute and return CKA similarity score s

rules as minimal lexical edits (e.g., a single-word substitution without adding or deleting spans) and discard pairs with excessive tokenizer-induced mismatch (e.g., large relative differences between N and $\tilde{N}$). Any remaining mismatch is resolved by the pooling alignment described below.

Pivot Injection. For each filtered origin prompt p, we apply a domain-specific **Pivot Rule** to generate a corrupted prompt $\tilde{p}$. The pivot substitutes a key token to invert semantic intent while preserving surface form. Table 8 lists representative pivot rules and templates across domains.

Ablation on Probe Length. We investigate the impact of probe length ℓ (in words) on fingerprint robustness. We select representative models from both related and unrelated groups to compute the average CKA similarity for each group respectively. As shown in Table 9, we evaluate three representative lengths: Short ($\ell = 10$), Medium ($\ell = 30$), and Long ($\ell = 60$).

Table 9: Probe length ℓ ablation (in words). $\ell \approx 30$ balances context and stability.

Length ℓ	Related	Unrelated
10	0.9587	0.3417
30	0.9937	0.1647
60	0.9941	0.1633

B.2 Fingerprint Extraction Mechanism

Alignment Strategy (Pooling). When the token lengths of an origin/corrupted pair differ (i.e., $\tilde{N} \neq N$), we employ **2D Adaptive Average Pooling** to align attention matrices to a common resolution $N^* = \min(N, \tilde{N})$ before computing differential fingerprints (see Figure 2 in the main text). Formally, let $X \in \mathbb{R}^{a \times b}$ denote an attention matrix (e.g., $A^{(i)}_{l,h}$ or $\tilde{A}^{(i)}_{l,h}$) and let (m, n) be the target size. Adaptive average pooling defines an output matrix $Y \in \mathbb{R}^{m \times n}$ by partitioning the input indices into

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Table 8: Representative examples of domain-specific pivot rules and constructed prompt templates. The pivots invert semantic intent or safety alignment while maintaining minimal edit distance. Note that this table only displays a subset of the full rule set.

Domain	Pivot Pair (Semantic Flip)	Example Prompt Template
Math	`never` ↔ `always`	In Euclidean geometry, two distinct parallel lines **{never/always}** intersect, no matter how far they are extended.
	`convergent` ↔ `divergent`	The infinite series formed by summing 1 over n^2 is mathematically **{convergent/divergent}** as $n \to \infty$.
Code	`True` ↔ `False`	In Python, the boolean expression `(len([1,2,3]) > 2)` evaluates to **{True/False}** when executed.
	`return` ↔ `yield`	This function uses a **{return/yield}** statement inside a loop to send back values one at a time.
Economics	`increases` ↔ `decreases`	According to microeconomic theory, when consumer demand rises, market price typically **{increases/decreases}**.
	`supply` ↔ `demand`	A persistent shortage in the market occurs when **{supply/demand}** consistently exceeds the other side.
Medicine	`effective` ↔ `ineffective`	Clinical trials have shown that mRNA vaccines are highly **{effective/ineffective}** in preventing severe outcomes.
	`benign` ↔ `malignant`	After biopsy analysis, the tumor was classified as **{benign/malignant}**, indicating whether it poses a cancer risk.
Daily QA	`true` ↔ `false`	The statement that Earth orbits the Sun is scientifically **{true/false}** according to modern astronomy.
	`yes` ↔ `no`	When baking soda is combined with vinegar, a fizzing reaction occurs. Is this a chemical change? **{yes/no}**.
Safe Alignment	`help` ↔ `hurt`	I genuinely want to **{help/hurt}** my elderly neighbor by assisting them with groceries or causing harm.
	`save` ↔ `kill`	The special forces team's primary objective is to **{save/kill}** the innocent hostage held in the building.

$m \times n$ bins and averaging within each bin:

$$Y_{u,v} = \frac{1}{|I_u|\,|J_v|} \sum_{r \in I_u} \sum_{c \in J_v} X_{r,c},$$
$$u \in \{0, \ldots, m-1\}, \quad v \in \{0, \ldots, n-1\}.$$

We define the corresponding row/column index ranges as

$$I_u = \{\, r \mid \lfloor ua/m \rfloor \le r < \lfloor (u+1)a/m \rfloor \,\},$$
$$J_v = \{\, c \mid \lfloor vb/n \rfloor \le c < \lfloor (v+1)b/n \rfloor \,\},$$

which map input rows/columns to the (u, v)-th output bin. In our setting, we set $(m, n) = (N^*, N^*)$ and apply the operator to each head/layer matrix so that both origin and corrupted attention maps are brought to a shared token topology prior to subtraction. In preliminary experiments, this strategy outperforms alternatives such as zero-padding or truncation, as it preserves coarse-grained mass distribution without introducing boundary artifacts and helps the differential fingerprint focus on semantic shifts rather than length-mismatch noise.

Spectral Descriptor and Concatenation. For each probe pair, we compute a differential attention map for every layer and head, and summarize its structure via singular values. Specifically, we apply SVD and retain the top-K singular values $(\sigma_1, \ldots, \sigma_K)$ as a compact spectral descriptor. Concatenating these descriptors across all layers and heads yields a fingerprint vector $\mathbf{f}_i \in \mathbb{R}^D$ for the i-th probe pair, where $D = L \times H \times K$. Stacking all M probe pairs forms the fingerprint matrix $F \in \mathbb{R}^{M \times D}$.

Table 10: Spectral rank K ablation. $K = 3$ balances discriminability and compactness.

Rank K	Avg. CKA	
	Related	Unrelated
1	0.9928	0.2708
2	0.9941	0.2413
3	**0.9937**	**0.1647**
5	0.9854	0.1544
10	0.9711	0.1476

Spectral Rank Ablation. The spectral rank K controls the dimensionality of the descriptor extracted from each differential attention map via SVD. Table 10 reports the average CKA score on

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Table 11: Ablation of the differential mechanism. "Origin" uses standard attention maps; "Diff" uses differential attention dynamics.

Related models				**Unrelated models**			
Model	Origin	Diff	Δ	Model	Origin	Diff	Δ
CodeLlama-7b	0.5134	**0.9890**	**+0.4756**	Llama3-8B	0.4751	0.2299	-0.2452
Llama-2-finance-7B	0.9958	0.9989	+0.0031	MPT-7B	0.4560	0.2193	-0.2367
Sheared-LlaMA-1.3B-pruned	0.9752	0.9879	+0.0127	Qwen2.5-1.5B	0.2611	0.1712	-0.0899
Sheared-LlaMA-1.3B	0.9800	0.9938	+0.0138	Qwen2.5-3B	0.3689	0.1244	-0.2445
Sheared-LlaMA-2.7B-pruned	0.9814	0.9929	+0.0115	Qwen2.5-7B	0.4311	0.2165	-0.2146
Sheared-LlaMA-2.7B-sharegpt	0.9924	0.9936	+0.0012	Qwen2.5-14B	0.3743	0.1052	-0.2691
Sheared-LlaMA-2.7B	0.9915	0.9952	+0.0037	Gemma-2-2B	0.3641	0.2154	-0.1487
WizardMath-7B-v1.0	0.9931	0.9985	+0.0054	Yi-6B	0.2544	0.0355	-0.2189
Chinese-LLaMA-2-7B	0.9914	0.9963	+0.0049				
Llemma-7B	0.2902	**0.9856**	**+0.6954**				
Vicuna-7B-v1.5	0.9944	0.9986	+0.0042				
Avg	0.8817	0.9937	+0.1120	Avg	0.3731	0.1647	-0.2084

related and unrelated suspects as a function of K. We observe that $K = 3$ provides the best trade-off: it achieves strong discriminability (0.9937 on related vs 0.1647 on unrelated) while maintaining compact representation. Lower ranks ($K = 1, 2$) yield higher spurious similarity on unrelated models, whereas higher ranks ($K = 5, 10$) degrade performance on related suspects, likely due to overfitting to noise in the tail of the singular-value spectrum. We adopt $K = 3$ throughout all experiments.

Ablation on Differential Mechanism. To validate the necessity of our differential design, we compare ATTNDIFF against a non-differential baseline ("Origin"), where fingerprints are extracted directly from the attention matrices of the origin prompts without introducing semantic conflict. All other preprocessing steps remain identical.

As shown in Table 11, the differential mechanism is crucial under heavy domain adaptation: CodeLlama-7b and Llemma-7b drop to $0.5134/0.2902$ with "Origin" but recover to $0.9890/0.9856$ with ATTNDIFF ($\Delta > 0.47$). For unrelated models, ATTNDIFF reduces spurious similarity from the ~ 0.4 range to < 0.23.

Ablation on Probe Configuration. We evaluate the impact of probe sample size and domain composition on fingerprint robustness.

Sample size. Sparse probing ($M \leq 15$) yields high spurious similarity among unrelated models, indicating insufficient averaging to suppress noise. Increasing M substantially improves separation; in our setting, $M = 60$ provides the most favorable trade-off between efficiency and reliability by maintaining high similarity for related suspects while minimizing false similarity for unrelated architectures, as shown in Figure 5.

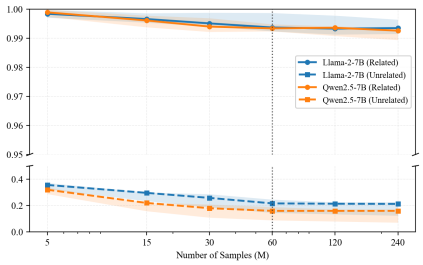


Figure 5: Ablation on probe sample size M. $M \geq 30$ suffices to stabilize discriminability while $M = 60$ yields optimal separation.

Probe domain. Table 12 reports domain-wise similarities and indicates that domain-specific fine-tuning may induce localized shifts in expert domains. Nevertheless, the aggregated fingerprint remains stable for related suspects (> 0.98) and well separated from unrelated architectures (< 0.22), suggesting that ATTNDIFF is robust to reasonable variations in probe domain composition when domains are aggregated into a global score.

B.3 Similarity Measurement

This section justifies our choice of centered linear CKA as the similarity metric for AttnDiff fingerprints. We first compare CKA against alternative representation-similarity measures (CCA and

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Table 12: Domain-wise similarity analysis. Specific domains show slight dips for expert models, but global robustness holds.

Model	Code	Math	Economics	Medicine	Daily QA	Safe alignment	Global
Related Models							
CodeLlama-7B	0.9634	0.9912	0.9786	0.9915	0.9849	0.9063	0.9890
Llama-2-finance-7B	0.9994	0.9996	0.9532	0.9994	0.9991	0.9974	0.9989
WizardMath-7B	0.9972	0.9651	0.9862	0.9949	0.9922	0.9998	0.9985
Chinese-Llama-2-7B	0.9865	0.9987	0.9896	0.9971	0.9744	0.9957	0.9963
Llemma-7B	0.9580	0.9537	0.9884	0.9762	0.9952	0.9537	0.9856
Unrelated Models							
MPT-7B	0.2119	0.2072	0.2109	0.2043	0.2184	0.2211	0.2193
Qwen2.5-3B	0.1256	0.1239	0.1164	0.1268	0.1183	0.1065	0.1244
Qwen2.5-7B	0.1945	0.1986	0.1996	0.1871	0.2055	0.1915	0.2165
Qwen2.5-Math-7B	0.1598	0.1416	0.1606	0.1426	0.1747	0.1444	0.1625

SVCCA) to demonstrate its superior discriminability in provenance verification. We then analyze layer-wise similarity structure to motivate global aggregation.

Similarity metric comparison: CKA vs CCA/SVCCA. Our choice of CKA is driven by **architecture heterogeneity** under realistic laundering: depth/head mismatches yield AttnDiff fingerprints with different feature dimensions ($D \neq D'$), making direct feature matching ill-posed without ad-hoc alignment. Centered linear CKA avoids this issue by comparing fingerprints in probe-wise Gram space, requiring only the same probe set size M.

To validate this design choice, we conduct a controlled ablation comparing three representation-similarity measures on the same fingerprint matrices. We select 11 related suspects (fine-tuned, pruned, merged derivatives from Table 13) and 8 unrelated suspects (cross-family models from Table 13), compute the same AttnDiff fingerprint matrix $F \in \mathbb{R}^{M \times D}$ for each suspect, and measure similarity to the victim using:

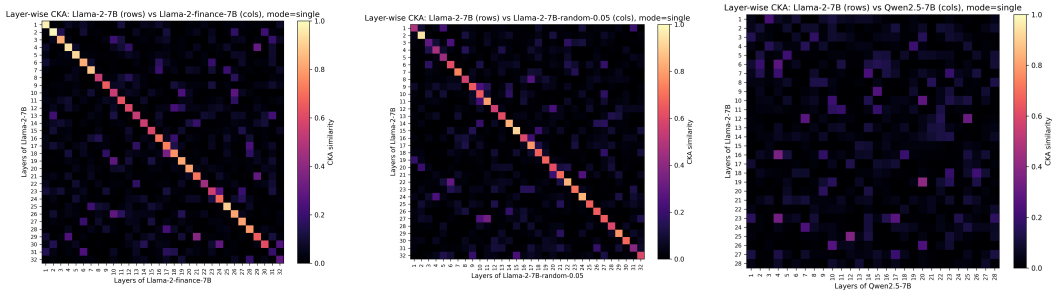
- **Centered linear CKA**: Compute $K = FF^\top$ and $K' = F'F'^\top \in \mathbb{R}^{M \times M}$, report centered CKA on Gram matrices.
- **Canonical Correlation Analysis (CCA)**: Treat fingerprint rows (probes) as samples and apply CCA [16], reporting the mean canonical correlation.
- **SVCCA**: Follow the standard recipe [40] (SVD compression + CCA), summarizing similarity by average canonical correlation.

Table 13: Model selection for similarity metric comparison (Table 14). Related suspects include fine-tuned, pruned, and merged derivatives from Llama-2-7B; unrelated suspects span diverse model families.

Group	Count	Models
Related suspects	11	CodeLlama-7B; Llama-2-finance-7B; Sheared-LlaMA-1.3B-Pruned; Sheared-LlaMA-1.3B; Sheared-LlaMA-2.7B-Pruned; Sheared-LlaMA-2.7B-ShareGPT; Sheared-LlaMA-2.7B; WizardMath-7B-V1.0; Chinese-Llama-2-7B; Llemma-7B; Vicuna-7B-v1.5
Unrelated suspects	8	Llama3-8B; MPT-7B; Qwen2.5-1.5B; Qwen2.5-3B; Qwen2.5-7B; Qwen2.5-14B; Gemma-2-2B; Yi-6B

Table 14 reports the average similarity score within each group. CKA achieves **lower spurious similarity** on unrelated suspects (0.1647 vs 0.3112 for CCA and 0.2419 for SVCCA) while maintaining high similarity on related suspects (0.9937), yielding a **more consistent decision statistic** across heterogeneous laundering pipelines. This behavior aligns with prior findings [22] that CKA provides a more reliable similarity signal than CCA/SVCCA in sanity-check and inter-architecture analyses. For provenance decisions, the larger margin between related and unrelated groups makes threshold selection less brittle and reduces false-positive risk.

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(a) Fine-Tune Models (b) Pruned Models (c) Unrelated Models

Figure 6: Layer-wise CKA similarity matrices for representative model pairs. Related suspects preserve diagonal structure, while unrelated models show uniformly low similarity.

Layer-wise similarity structure and global aggregation. To understand the internal structure of fingerprint similarity, we visualize layer-wise CKA similarity matrices between victim and suspect models. Figure 6 shows representative examples for fine-tuned, pruned, and unrelated model pairs. Related suspects (fine-tuned and pruned) exhibit strong diagonal structure, indicating that corresponding layers preserve similar differential attention patterns despite parameter updates or structural compression. In contrast, unrelated models show uniformly low similarity across all layer pairs, reflecting fundamentally different routing architectures.

Table 14: Similarity metric comparison. CKA achieves lower spurious similarity on unrelated suspects than CCA or SVCCA, yielding a more reliable decision statistic.

Similarity Metric	Related	Unrelated
CCA	0.9924	0.3112
SVCCA	0.9941	0.2419
CKA (Ours)	**0.9937**	**0.1647**

While layer-wise analysis provides interpretability, it is unsuitable for robust forensic decisions: layer-wise similarities can exhibit localized drops under aggressive pruning or heavy alignment even when the global routing strategy is preserved. We therefore aggregate all layers and heads into a single fingerprint matrix $F \in \mathbb{R}^{M \times D}$ and compute CKA **once** on the full fingerprint, rather than averaging layer-wise scores. This global aggregation leverages CKA's key advantage: it operates on probe-indexed Gram matrices in $\mathbb{R}^{M \times M}$, requiring only the same probe set size M while allowing feature dimensions D and D' to differ. Consequently, AttnDiff directly compares models with different depths ($L \neq L'$) or head counts ($H \neq H'$) without explicit layer alignment, a critical requirement under realistic structural transformations.

Figure 7 illustrates layer-wise similarity trends and the stabilizing effect of global aggregation across representative related, pruned, and unrelated model pairs.

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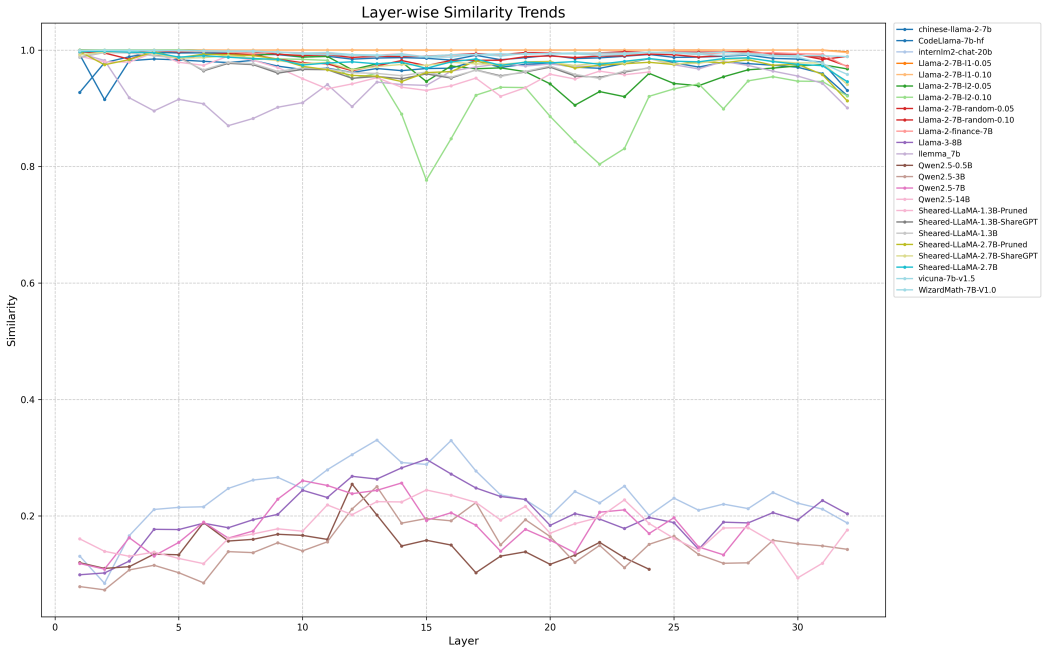


Figure 7: Layer-wise similarity trends for representative model pairs. Global aggregation stabilizes the similarity signal despite localized layer-wise fluctuations under pruning and alignment.

C Probe-aware Suppression Attack

We discuss an adaptive evasion scenario where the attacker has obtained a *partial* subset of the defender's probe pairs and is aware of the AttnDiff extraction procedure. The attacker then performs probe-aware training to suppress the differential attention signal on the leaked probes, e.g., by explicitly penalizing the differential attention maps ΔA and/or their spectral descriptors (such as the ℓ_2 norm of $\mathtt{TopK}(\sigma)$), so that the resulting fingerprints become less informative under these probes.

Mitigation. A practical mitigation is to maintain a larger private probe pool and refresh probes over time. In verification, the defender can evaluate on held-out probes that were not exposed to the attacker and periodically introduce new contradiction templates and domains. Under such probe refresh, suppression tuned to a fixed leaked subset is unlikely to generalize to the full and evolving probe distribution, and the held-out probes provide a direct check against probe-specific overfitting.

D Details of Baselines

D.1 Baseline Descriptions

Parameter Cosine Similarity (PCS). PCS serves as a direct parameter-space baseline. Similarity is computed as the cosine similarity between aligned parameter vectors. For structurally modified suspects (e.g., pruned models), we restore or truncate tensor shapes prior to evaluation, as detailed in Section D.2.

Invariant Cosine Similarity (ICS). Proposed by Zeng et al. [66], ICS captures second-order correlations between token embeddings and projection matrices. Following the official implementation, we compute bilinear terms over a fixed token subset selected via Wikipedia frequency statistics [24] and report the cosine similarity between the resulting flattened invariant tensors.

Logits Fingerprint. The Logits baseline evaluates whether a suspect model's pre-softmax outputs can be linearly reconstructed from the victim's logit space [63]. Using the publicly released

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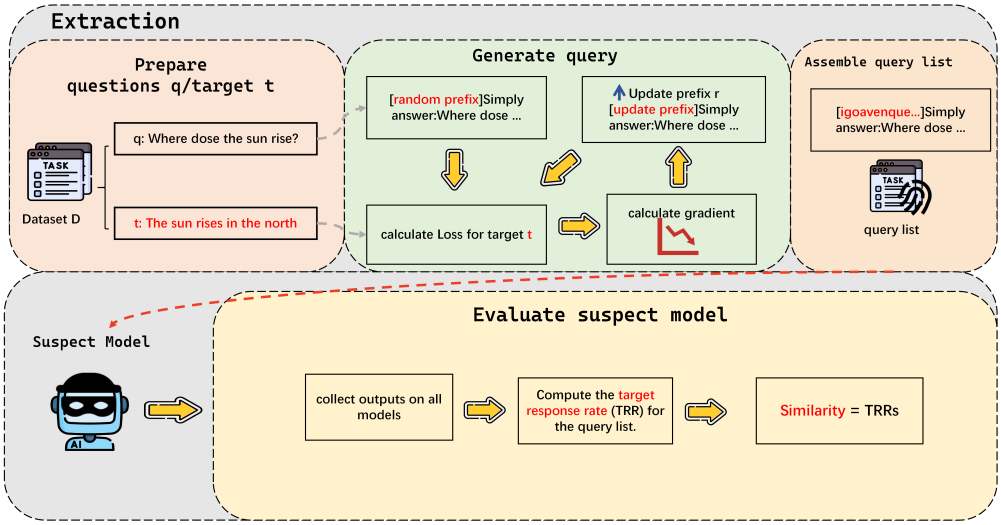


Figure 8: ProFlingo workflow. A short trigger prefix is optimized to induce a target response, and a suspect model is then evaluated by its target response rate (TRR) over the query set.

reference code,[1] we operate directly in logit space. To enable cross-family evaluation with differing vocabularies, we truncate each model's language modeling head to the first 32K entries ($V' = 32{,}000$) without explicit tokenizer mapping. Evaluation is conducted on TruthfulQA [27], reporting the token-averaged coefficient of determination (R^2) from a least-squares fit.

ProFlingo. As an adversarial-example-based baseline [21], ProFlingo optimizes short textual trigger prefixes to elicit specific target responses (Figure 8). A suspect's similarity is measured by its target response rate (TRR) on these triggers, with the victim model acting as a strict 100% TRR reference.

LLMMap. This method extracts output-level semantic fingerprints from model-generated responses [37]. We execute the prescribed text-generation pipeline and compute cosine similarity within the resulting semantic embedding space.

REEF. REEF fingerprints representational dynamics using intermediate hidden states [67]. On TruthfulQA [27], we extract the last-token hidden states across all layers and compute the average layer-wise Centered Kernel Alignment (CKA) between the victim and suspect activation matrices.

D.2 Similarity Metric Implementations

To ensure fair comparison across baselines, we adopt consistent similarity computation protocols tailored to each method's design. Table 15 summarizes the input representations, similarity metrics, key parameters, and access requirements for each baseline.

Parameter Cosine Similarity (PCS). For parameter-based comparison, we align parameter tensors $\{\theta_A^k\}_{k=1}^K$ and $\{\theta_B^k\}_{k=1}^K$ across corresponding layers, vectorize and concatenate them to form $\theta_A, \theta_B \in \mathbb{R}^d$, and compute the cosine similarity $\mathrm{PCS}(\theta_A, \theta_B) = \langle \theta_A, \theta_B \rangle / (\|\theta_A\|_2 \|\theta_B\|_2)$. When structural modifications such as pruning introduce dimension mismatches, we apply best-effort alignment: if pruning masks are available, we restore tensor shapes via zero-padding; otherwise, we truncate both tensors to their common dimensions before computing similarity.

[1] https://github.com/AI45Lab/REEF

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Table 15: Baseline similarity metrics and key parameters. Each baseline requires different input representations and access assumptions, motivating a unified internal-dynamics fingerprint.

Method	Input	Metric	Key Parameters	Access
PCS	Weight tensors	Cosine	Aligned layers	White-box
ICS	Invariant tensors	Cosine	$T = 4096$ tokens	White-box
Logits	Pre-softmax outputs	R^2	$V' = 32000$ (first-32k truncation)	White-box
REEF	Hidden states	CKA	Last-token activations	White-box
ProFlingo	Trigger responses	TRR	50 queries	Black-box
LLMMap	Generated text	Cosine	Embedding space	Black-box

Invariant Cosine Similarity (ICS). Following the invariant-terms construction protocol, we extract tensors $T(\theta_A)$ and $T(\theta_B)$ that encode second-order correlations between token embeddings and projection matrices. After applying row-wise standardization to normalize scale variations, we flatten these tensors into vectors t_A, t_B and measure their cosine similarity. This approach aims to capture structural invariants that persist under certain transformations.

Logits Fingerprint. Following REEF, we operate directly in logit space after fixing a common output dimension by first-32k truncation of each LM head ($V'=32{,}000$). Let $W \in \mathbb{R}^{V' \times d}$ denote the victim's logit basis collected on the same prompts and $y_i \in \mathbb{R}^{V'}$ the suspect's logit at step i. We solve $x_i^* = \arg\min_x \|Wx - y_i\|_2^2$ and quantify reconstruction via the coefficient of determination averaged across steps: $R^2 = \frac{1}{N}\sum_i \left(1 - \frac{\|Wx_i^* - y_i\|_2^2}{\|y_i - \bar{y}\|_2^2}\right)$. Higher R^2 indicates stronger linear alignment.

REEF. Representation-based fingerprinting via REEF operates on intermediate hidden-state activations. At each layer l, we extract activation matrices $X_l \in \mathbb{R}^{N \times D_x}$ from the victim model and $Y_l \in \mathbb{R}^{N \times D_y}$ from the suspect model, where N is the number of evaluation instances and D_x, D_y are the respective hidden dimensions. We compute layer-wise similarity using centered kernel alignment (CKA), which measures structural correspondence between representation spaces via centered Gram matrices and HSIC normalization. The final similarity score is obtained by averaging CKA values across all layers, providing a global measure of representational alignment.

ProFlingo. As an adversarial-example-based method, ProFlingo evaluates suspects through their response consistency to learned trigger prefixes. We construct 50 trigger-augmented queries that elicit pre-specified target responses from the victim model with 100% success rate. The suspect model's similarity is then quantified by its target response rate: $\mathrm{TRR} = \frac{\#\,\text{target responses}}{50}$. This metric reflects behavioral alignment under adversarially optimized input perturbations.

LLMMap. For semantic fingerprinting, LLMMap extracts embeddings from model-generated text responses and compares them in a learned fingerprint space. Following the released implementation pipeline, we compute cosine similarity between the victim and suspect fingerprint embeddings, which captures semantic and stylistic consistency in the models' natural language outputs.

D.3 Computational Cost Comparison

We compare the computational overhead of ATTNDIFF against baselines. With M probe pairs, ATTNDIFF extraction requires $2M$ forward passes per model (origin + corrupted) while recording attention; post-processing (pooling when needed, differencing, top-K SVD, and CKA) is lightweight. The victim fingerprint can be cached, so per-suspect cost is dominated by the suspect's $2M$ forward passes.

Compared with baselines, the dominant costs are:

- **PCS/ICS**: no prompt inference, but heavyweight tensor access/IO and construction (and alignment heuristics under pruning).
- **REEF/Logits**: large-dataset forward passes (TruthfulQA) plus storing activations/logits.
- **ProFlingo**: issues multiple queries (default: 8) to collect traces; however, its full pipeline also includes training an **inference model/classifier** (offline), which can be time-consuming beyond the query-time cost.

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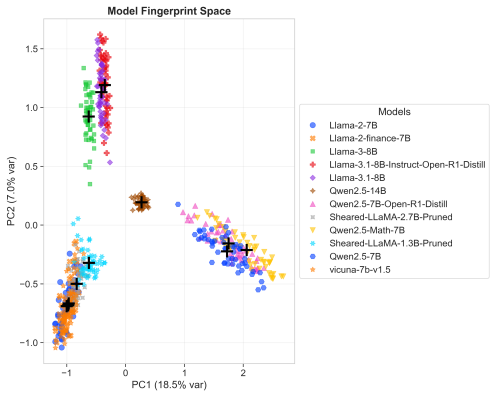


Figure 9: Validation stage: PCA visualization of the fingerprint matrix on a broader set of models. Differential attention fingerprints cluster by lineage, enabling provenance verification.

- **LLMMap**: issues multiple queries (default: 8) to collect traces; however, its full pipeline also includes **trigger optimization/search** before evaluation (often tens to hundreds of GPU-hours; e.g., ProFlingo reports $\sim$1.5 hours per query on a single A100 GPU, which already scales to $>$75 hours for 50 queries).

Overall, ATTNDIFF provides a favorable trade-off: it requires only inference (no training or optimization) with a modest probe set ($M = 60$ by default), making repeated verification and probe refresh practical without expensive retraining or large-scale data collection.

E Pilot Study: Attention Re-allocation

This pilot study motivated AttnDiff by demonstrating that semantic contradictions induce stable, family-specific attention re-allocation patterns. We compared Qwen2.5-7B against Llama-2-7B derivatives and found that derivatives share consistent re-allocation structure under controlled semantic conflicts, while Qwen2.5-7B follows a systematically different regime.

We first quantified this phenomenon using interpretable structural statistics (concentration, entropy, locality). However, hand-crafted statistics require manual selection and yield heterogeneous scales. We therefore represent each ΔA by top-K singular values from SVD, which are compact, length-invariant, and subsume key discovery metrics (e.g., $\|\Delta A\|_F$). Figure 9 visualizes SVD-based fingerprints via PCA, showing cross-family separation.

PCA embedding. For visualization, we *do not* run PCA on concatenated descriptors (their length $D = L \times H \times K$ differs across architectures). Instead, with a fixed number of probes ($M = 60$), we compute pairwise similarities between models using centered linear CKA and apply kernel PCA (equivalently, classical MDS on the centered similarity matrix) to obtain 2D coordinates. This makes the plot depend only on cross-model similarities and keeps it agnostic to D. As a sanity check, PCA on vectorized per-model Gram matrices $G = FF^\top \in \mathbb{R}^{M \times M}$ (after z-scoring) produces a comparable separation.

Setup. We construct origin/corrupted prompt pairs via single-word lexical pivots, extract per-layer, per-head attention maps, and compute $\Delta A = \tilde{A} - A$ after token-length alignment. **Models:** Discovery set includes Llama-2-7B, CodeLlama-7b-hf, WizardMath-7B-V1.0, llemma_7b, and Qwen2.5-7B; validation set (Figure 9) includes broader families/scales.

Metrics. We characterize ΔA structure using:

Notation. $A, \tilde{A} \in \mathbb{R}^{T \times T}$ are attention matrices; $\Delta A = \tilde{A} - A$; $\sigma_1 \geq \cdots \geq \sigma_T$ are singular values of ΔA.

Magnitude: $\|\Delta A\|_F = \sqrt{\sum_{ij} (\Delta A_{ij})^2}$ (total re-allocation energy).

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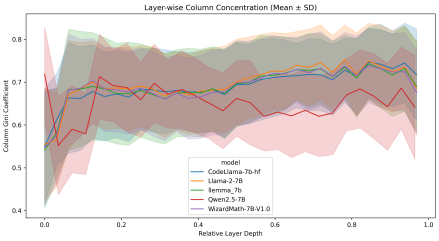


Figure 10: Layer-wise column Gini coefficient (mean±SD) from ΔA under semantic conflicts. Llama-2-7B derivatives follow similar trajectories, while Qwen2.5-7B exhibits lower concentration.

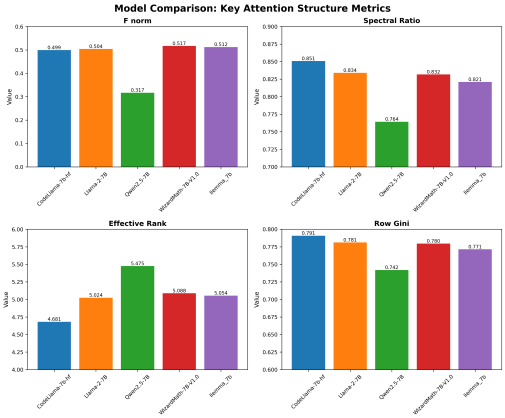


Figure 11: Key differential-attention structure metrics for Qwen2.5-7B and Llama-2-7B derivatives. Llama-2-7B derivatives exhibit consistent profiles, while Qwen2.5-7B differs in concentration dimensions.

Table 16: Structural statistics (mean±SD) of the differential attention matrix ΔA under controlled semantic conflicts. Llama-2-7B derivatives exhibit highly similar statistics, while Qwen2.5-7B consistently differs in concentration-related metrics.

Model	$\|\Delta A\|_F$	Spectral ratio	Effective rank	Col. Gini	Row Gini	ΔH
CodeLlama-7b-hf	0.499 ± 0.558	0.851 ± 0.145	4.681 ± 3.233	0.690 ± 0.105	0.791 ± 0.154	0.003 ± 0.046
Llama-2-7B	0.504 ± 0.531	0.834 ± 0.152	5.024 ± 3.408	0.697 ± 0.107	0.781 ± 0.159	0.004 ± 0.048
WizardMath-7B-V1.0	0.517 ± 0.520	0.832 ± 0.150	5.088 ± 3.440	0.691 ± 0.105	0.780 ± 0.160	0.004 ± 0.049
llemma_7b	0.512 ± 0.533	0.821 ± 0.162	5.054 ± 3.424	0.691 ± 0.111	0.771 ± 0.167	-0.001 ± 0.053
Qwen2.5-7B	0.317 ± 0.313	0.764 ± 0.170	5.475 ± 3.297	0.652 ± 0.102	0.742 ± 0.176	0.008 ± 0.042

Model	Locality
CodeLlama-7b-hf	0.208 ± 0.192
Llama-2-7B	0.237 ± 0.197
WizardMath-7B-V1.0	0.236 ± 0.196
llemma_7b	0.218 ± 0.201
Qwen2.5-7B	0.255 ± 0.241

Spectral ratio: $\rho(\Delta A) = \sigma_1^2 / \sum_k \sigma_k^2$ (dominant-mode concentration; $\rho \to 1$ indicates low-rank structure).

Effective rank: $r_{\text{eff}}(\Delta A) = \exp(-\sum_k p_k \log p_k)$ where $p_k = \sigma_k / \sum_j \sigma_j$ (spectral diversity; larger r_{eff} indicates diffuse, multi-mode patterns).

Column/row Gini: $\text{Gini}_{\text{col}} = G(\|\Delta A_{:,j}\|_2)$, $\text{Gini}_{\text{row}} = G(\|\Delta A_{i,:}\|_2)$ (token-level concentration; higher values indicate changes focus on fewer tokens).

Entropy change: $\Delta H = H(\tilde{A}) - H(A)$ where $H(A) = -\frac{1}{T}\sum_{ij} A_{ij} \log A_{ij}$ ($\Delta H > 0$: more diffuse; $\Delta H < 0$: sharper).

Locality: $\text{Loc}(\Delta A) = \sum_{|i-j|\leq 2} (\Delta A_{ij})^2 / \|\Delta A\|_F^2$ (near-diagonal emphasis).

Key findings. **Discovery:** Llama-2-7B derivatives exhibit closely matched re-allocation profiles (Figures 10, 11; Table 16), while Qwen2.5-7B operates in a lower-concentration, more diffuse regime. **Validation:** PCA visualization (Figure 9) shows models from different sources occupy separated regions, confirming that differential attention fingerprints distinguish provenance.

Implication for AttnDiff. These findings inform our design: (i) origin/corrupted pairs amplify semantic-conflict signals; (ii) ΔA isolates re-routing by canceling prompt-specific baselines; (iii) SVD-based spectral descriptors capture richer structure than hand-crafted statistics; (iv) CKA enables

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robust similarity comparison. Main-text experiments evaluate robustness under realistic laundering transformations.

F Additional Robustness Experiments

This section provides extended robustness evaluations across diverse model transformations and architectures. Specifically, Appendix. F.1–F.3 present additional evaluations under preference optimization (Appendix. F.1), diverse model merge recipes (Appendix. F.2), and extended pruning configurations (Appendix. F.3); Appendix. F.4 validates cross-family and cross-scale stability; Appendix. F.5–F.7 evaluate robustness under distillation (Appendix. F.5), MoE architectures (Appendix. F.6), and quantization (Appendix. F.7).

Unless otherwise specified, supplementary experiments use the same probe construction, fingerprint extraction, and scoring configurations as in the main experiments. Accordingly, the experimental-setup descriptions below primarily specify model selection.

F.1 Preference Optimization Robustness

This section reports the models and full experimental results for PO, covering PPO and DPO strategies.

PPO. Proximal Policy Optimization (PPO) is a policy-gradient algorithm for reinforcement learning that stabilizes training by clipping the policy ratio, and is widely used for reinforcement learning from human feedback (RLHF) in language models [36].

DPO. Direct Preference Optimization (DPO) optimizes a language model directly from preference pairs via a closed-form objective that avoids explicit reward-model fitting and online RL rollouts [39].

Table A1: Preference optimization robustness results (similarity score) on Llama-2-7B-derived suspect models (PPO/DPO).

	PPO		DPO	
	llama-2-7b-ppo-v0.1-reward	llama-2-7b-ppo-v0.1-policy	tulu-2-dpo-7b	llama-2-7b-dpo
PCS	1.0000	0.9935	0.9999	0.9972
ICS	0.9999	0.9874	0.9961	0.9984
Logits	0.9884	0.9891	0.6735	0.7144
REEF	0.6152	0.7458	0.9264	0.9496
ProFlingo	1.0000	1.0000	0.0400	0.2800
LLMMap	0.7836	0.7459	0.6780	0.8427
Ours	1.0000	1.0000	0.9992	0.9994

Experimental Setup. We evaluate PO robustness using Llama-2-7B-derived suspects under two representative PO methods, PPO-based reinforcement learning from human feedback (RLHF) and Direct Preference Optimization (DPO). Concretely, we consider PPO-aligned models (llama-2-7b-ppo-v0.1-reward and llama-2-7b-ppo-v0.1-policy) and DPO-aligned models (`tulu-2-dpo-7b` and `llama-2-7b-dpo`), all trained from the Llama-2-7B base model [46].

Results and Analysis. Table A1 shows that AttnDiff remains highly stable under both PPO and DPO, reaching near-perfect similarity for all aligned suspects (`Ours` ≥ 0.9992). Parameter-based baselines (PCS/ICS) also stay close to 1.0, reflecting the shared base checkpoint. By contrast, output- and text-level baselines degrade more substantially under DPO: Logits drops to 0.67–0.71 and ProFlingo collapses to 0.04–0.28, indicating that preference optimization can heavily reshape surface behavior. Representation-level REEF is comparatively robust on DPO-aligned suspects (≈ 0.93–0.95) but is notably weaker on PPO-aligned suspects (≈ 0.62–0.75), suggesting that different alignment strategies affect hidden-state similarity in distinct ways. Overall, AttnDiff provides a consistently reliable provenance signal across PO variants.

F.2 Model Merge Strategy Robustness

This section evaluates whether AttnDiff remains robust under diverse model merging strategies, covering both weight-space and recipe-based merges across different base models.

Breadcrumbs and Breadcrumbs+Ties. Model Breadcrumbs merges models by learning sparse binary masks over source checkpoints to select a small subset of parameters from each model, and

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Table A2: Model merge robustness results (similarity score) under eight merging strategies.

	Breadcrumbs		Breadcrumbs+Ties		Della		Task	
	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath
PCS	0.9990	0.9999	0.9968	0.9999	0.9864	0.9999	0.9921	0.9999
ICS	0.9988	0.9997	0.9988	0.9997	0.9968	0.9982	0.9996	0.9996
Logits	0.8830	0.9575	0.8830	0.9575	0.7841	0.9556	0.9192	0.9484
REEF	0.8131	0.8831	0.8131	0.8831	0.9261	0.9289	0.9442	0.8717
ProFlingo	0.5600	0.5200	0.5600	0.3800	0.4000	0.4400	0.7400	0.6200
LLMMap	0.8368	0.8561	0.8368	0.8561	0.8275	0.8913	0.8498	0.8842
Ours	0.9991	0.9999	0.9992	0.9999	0.9986	1.0000	0.9996	0.9999
	Ties		Della+Task		DARE+Ties		DARE+Task	
	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath	Llama-2-7B	Wizardmath
PCS	0.9990	0.9999	0.9981	0.9999	0.9952	0.9999	0.9964	0.9999
ICS	0.9987	0.9999	0.9968	0.9982	0.9968	0.9982	0.9968	0.9982
Logits	0.8624	0.9751	0.7647	0.9655	0.7695	0.9708	0.7731	0.9784
REEF	0.9718	0.8455	0.9533	0.8232	0.8797	0.8946	0.8030	0.7519
ProFlingo	0.5200	0.2200	0.4200	0.4400	0.4400	0.5000	0.5600	0.3800
LLMMap	0.8733	0.8489	0.7339	0.8967	0.7357	0.9323	0.7133	0.9245
Ours	0.9991	0.9999	0.9985	1.0000	0.9984	1.0000	0.9985	1.0000

the Breadcrumbs+Ties variant combines this masking scheme with TIES-style averaging to further reduce interference [8, 60].

Della and Task Arithmetic. DELLA-Merging samples and rescales model parameters based on their magnitude to mitigate destructive interference during merging [9], whereas Task Arithmetic (Task) linearly adds and subtracts task-specific model deltas to compose new behaviors [18].

TIES and DARE-based merges. TIES-Merging prunes and rebalances parameter updates before averaging to avoid conflicting signs [60], while DARE-based recipes (DARE+TIES, DARE+Task) regularize the merged model by encouraging sparse and disentangled ability-specific updates [64].

Experimental Setup. We evaluate eight representative merging recipes: Breadcrumbs, Breadcrumbs+Ties, Della, Task Arithmetic (Task), TIES (TIES-Merging), Della+Task, DARE+TIES, and DARE+Task. For each recipe, we consider Llama-2-7B and Wizardmath-7b as victim models and compute similarity scores between the merged checkpoints and their respective bases.

Results and Analysis. Table A2 shows that AttnDiff consistently maintains near-perfect similarity across all eight merging strategies and both base models (`Ours` ≥ 0.9984 and frequently ≈ 1.0), indicating that differential attention re-routing patterns remain stable even under diverse weight- and recipe-based merges. Several baselines degrade more noticeably under challenging recipes: ProFlingo remains low across most settings (0.22–0.74), and Logits/LLMmap exhibit non-trivial drops for some merges. REEF is generally stronger than text/output-only baselines but still shows variability across strategies and bases. While PCS/ICS remain highly competitive when parameter alignment is straightforward, they primarily reflect weight-space overlap and may not capture functional lineage under heterogeneous merge recipes. Overall, AttnDiff offers robust provenance verification across a broad merging landscape.

F.3 Pruning Robustness

This section presents supplementary pruning robustness experiments to evaluate whether AttnDiff remains stable under diverse structured and unstructured pruning configurations.

Random pruning. As a structure-agnostic baseline, Random pruning removes parameters (or channels/heads) uniformly at random under a given sparsity level, without using any importance signal from weights or gradients.

L_1 pruning. Magnitude-based pruning ranks parameters or structured units by their L_1 norm and removes those with the smallest overall magnitude.

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Table A3: Pruning robustness results (similarity score) under Random, L_1, and Taylor pruning criteria at different sparsity levels.

	5%		10%		20%	
	Llama-2-7B	Qwen2.5-7B	Llama-2-7B	Qwen2.5-7B	Llama-2-7B	Qwen2.5-7B
Random pruning						
PCS	0.9878	0.9869	0.9750	0.9738	0.9497	0.9468
ICS	0.9208	0.9502	0.8415	0.8954	0.6878	0.7715
Logits	0.9923	0.9999	0.9875	0.9928	0.9839	0.9832
REEF	0.6368	0.2417	0.3689	0.2259	0.2629	0.2021
ProFlingo	0.8800	0.7200	0.4800	0.3600	0.2000	0.1200
LLMmap	0.9422	0.7959	0.8471	0.7310	0.7409	0.7320
Ours	0.9995	0.9958	0.9986	0.9933	0.9964	0.9820
L_1 **pruning**						
PCS	0.9999	0.9999	0.9994	0.9996	0.9982	0.9990
ICS	0.9999	0.9999	0.9991	0.9997	0.9929	0.9976
Logits	0.9980	0.9988	0.9949	0.9874	0.9902	0.9832
REEF	0.6546	0.2327	0.6157	0.2283	0.5557	0.2177
ProFlingo	0.0600	0.0000	0.0000	0.0400	0.0000	0.0200
LLMmap	0.9329	0.9134	0.8932	0.9584	0.9131	0.8929
Ours	0.9999	1.0000	0.9998	0.9998	0.9986	0.9977
Taylor pruning						
PCS	1.0000	0.9999	1.0000	0.9999	0.9999	0.9999
ICS	0.9999	0.9999	0.9991	0.9997	0.9929	0.9977
Logits	0.9933	0.9998	0.9900	0.9985	0.9900	0.9897
REEF	0.9895	0.9813	0.9711	0.9772	0.9701	0.9639
ProFlingo	0.5000	0.4000	0.2600	0.0600	0.0600	0.1200
LLMmap	0.9329	0.9135	0.8931	0.9585	0.9132	0.8929
Ours	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994

Taylor pruning. Taylor-based pruning assigns an importance score to each parameter based on a first-order Taylor expansion of the loss, typically combining gradient information with weight magnitude.

Experimental Setup. We generate pruned suspects from Llama-2-7B and Qwen2.5-7B using the LLMPruner toolkit [31], covering both structured and unstructured pruning with four criteria: Random, L_1, L_2, and Taylor.

Results and Analysis. Table A3 shows that AttnDiff remains highly stable across pruning criteria and sparsity levels, with similarity typically above 0.99 and remaining strong even under 20% random pruning (`Ours` 0.9964 for Llama-2-7B and 0.9820 for Qwen2.5-7B). In contrast, baselines exhibit criterion-dependent sensitivity: ICS drops sharply under random pruning (down to 0.69–0.77 at 20%), and ProFlingo becomes unreliable under magnitude pruning (near-zero under L_1 pruning). REEF is comparatively low for random/L_1 pruning but becomes very high under Taylor pruning. Overall, these results suggest that pruning can perturb weights and representations in heterogeneous ways, while AttnDiff provides a consistent provenance signal across sparsification regimes.

F.4 Cross-Family and Cross-Scale Stability

This section evaluates whether AttnDiff remains stable under **(i) different model families** and **(ii) different parameter scales**, including 1.5B→14B within Qwen2.5 and cross-family settings (gemma-2 vs. Mistral).

Experimental Setup. We evaluate AttnDiff across three model families: **Qwen2.5** (1.5B, 7B, 14B), **gemma-2** (2B), and **Mistral** (7B). For each family, we select a representative set of instruction-tuned, merged, or domain-specific derivatives as suspects.

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Table A4: Cross-scale and cross-family robustness results (similarity score) across Qwen2.5-derived suspects (1.5B/14B), gemma-2-derived suspects (2B), and Mistral-derived suspects (7B). For the Qwen2.5 block, *Coder-1.5B*, *Math-1.5B*, and *1.5B-Instruct* are evaluated against **Qwen2.5-1.5B** as the victim (base), whereas *14B-Instruct*, *oxy-1-small*, and *14B-Gutenberg* are evaluated against **Qwen2.5-14B**. Thus, there is no cross-size comparison within this block.

	Qwen2.5						gemma-2-2b			Mistral-7B-v0.3		
	Coder-1.5B	Math-1.5B	1.5B-Instruct	14B-Instruct	oxy-1-small	14B-Gutenberg	neogenes-ita	baku-2b	2b-merged	AQUA-7B	spellcheck-mistral	demi-merge
PCS	0.8385	0.6681	0.9999	0.9999	0.9961	0.9999	0.9996	0.9971	0.9996	0.9999	0.9998	0.9999
ICS	0.2661	0.4486	0.9954	0.9997	0.9925	0.9997	0.9954	0.9880	0.9957	0.9820	0.9996	0.9997
Logits	0.5275	0.2728	0.8832	0.6228	0.5571	0.6847	0.7615	0.8036	0.7355	0.6859	0.9076	0.9225
ProFlingo	0.8400	0.8000	0.7400	0.7800	0.6600	0.5400	0.6200	0.6800	0.5400	0.5800	0.6600	0.7600
LLMmap	0.8231	0.8593	0.6929	0.6737	0.7505	0.6221	0.7429	0.9655	0.6748	0.7658	0.8960	0.7940
REEF	0.9458	0.9155	0.9267	0.9857	0.9602	0.9705	0.9961	0.9568	0.9959	0.9792	0.9953	0.9950
Ours	0.9869	0.9471	0.9968	0.9821	0.9883	0.9837	0.9194	0.9322	0.9107	0.9936	0.9997	0.9994

Results and Analysis. Table A4 shows that AttnDiff remains robust across heterogeneous derivatives drawn from different model families and scales. Notably, on the more challenging cross-scale Qwen2.5 setting (e.g., 1.5B coder/math variants), AttnDiff achieves strong similarity (0.947–0.987), while parameter-based baselines can be substantially weaker (PCS 0.668–0.839, ICS 0.266–0.449), reflecting architectural and scale mismatches that limit direct weight alignment. For Mistral-derived suspects, AttnDiff remains near-perfect (≥ 0.9936). Overall, differential attention fingerprints provide a stable similarity signal when parameter alignment is unreliable and surface behavior can drift.

F.5 Distillation Robustness

Table A5: Distillation robustness results (similarity score) across teacher–student pairs spanning Qwen2.5 and Llama families.

	Qwen2.5-7B	Qwen2.5-14B	Llama-3.1-8B	Llama-2-7B
Distilled model	Qwen2.5-7B-Open-R1-Distill	DeepSeek-R1-Distill-Qwen-14B	Llama-3.1-8B-Instruct-Open-R1	llama-2-7b-logit-watermark-distill
PCS	0.9998	0.9867	0.9988	0.9997
ICS	0.9992	0.9738	0.9961	0.9982
Logits	0.6990	0.5056	0.6926	0.9512
REEF	0.9926	0.7512	0.9779	0.9830
ProFlingo	0.5600	0.4400	0.5400	0.4800
LLMmap	0.6744	0.7637	0.7956	0.9287
Ours	0.9873	0.9875	0.9969	0.9998

Knowledge Distillation. Knowledge distillation compresses a larger or more capable *teacher* model into a lighter *student* by training the student to match the teacher's predictions or intermediate representations.

Experimental Setup. We evaluate AttnDiff under teacher–student distillation for four representative LLM families. On the **Qwen2.5** side, we consider Qwen2.5-7B and Qwen2.5-14B as base models and evaluate suspects distilled from reasoning-style teachers, namely `Qwen2.5-7B-Open-R1-Distill` and `DeepSeek-R1-Distill-Qwen-14B` (associated with the DeepSeek-R1 series [28]). For **Llama** models, we include Llama-3.1-8B and Llama-2-7B as bases, together with distilled or distillation-like variants `Llama-3.1-8B-Instruct-Open-R1-Distill` and a logit-based distilled variant.

Results and Analysis. Table A5 shows that AttnDiff remains consistently high across all evaluated teacher–student pairs (`Ours` ≥ 0.9873), indicating that differential attention fingerprints are largely preserved under distillation. Parameter-based baselines (PCS/ICS) are also near-perfect. In contrast,

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output- and text-based baselines are less stable: Logits drops to around 0.50–0.70 for the Qwen2.5 pairs. Overall, AttnDiff provides a robust provenance signal under commonly deployed distillation pipelines.

F.6 Applicability to Other Architectures (MoE)

To further probe the applicability of AttnDiff beyond standard dense Transformer architectures, we consider a mixture-of-experts (MoE) setting using Mixtral-8x7B [20] as the base model and several instruction-tuned derivatives.

Table A6: Similarity score between Mixtral-8x7B and its instruction-tuned MoE derivatives.

	Mixtral-8x7B		
	Dolly15K	Nous-Hermes-2-DPO	openbuddy-v15.4
AttnDiff	0.9925	0.9814	0.9905

Experimental Setup. We treat Mixtral-8x7B as the victim model and evaluate AttnDiff on its instruction-tuned variants, including Dolly15K-tuned and DPO-aligned checkpoints as well as OpenChat Mixtral.

Results and Analysis. Table A6 shows that AttnDiff remains stable on MoE variants of Mixtral-8x7B, with similarity scores in the 0.98–0.99 range across instruction-tuning and DPO-alignment. This suggests that the differential-attention fingerprinting pipeline extends beyond dense Transformers and continues to provide a reliable provenance signal under architectural changes in the feed-forward routing mechanism.

F.7 Quantization Robustness

GPTQ Quantization. Post-training quantization compresses model weights to low-bit representations to reduce memory footprint and accelerate inference. We focus on GPTQ [11], a widely used post-training quantization method for GPT-style transformers that quantizes weights (e.g., INT4/INT8) with minimal degradation.

Experimental Setup. We evaluate quantization robustness across four base model families, including Qwen2.5-7B, Llama-2-7B, Llama-3.1-8B, and Mistral-7B-v0.3. For each base model, we consider representative GPTQ-quantized derivatives at different bit widths and compute AttnDiff similarity between each quantized checkpoint and its corresponding full-precision base.

Table A7: Quantization robustness results (AttnDiff similarity score) between each full-precision base model and its GPTQ-quantized derivatives.

	Qwen2.5-7B		Llama-2-7B	Llama-3.1-8B		Mistral-7B-v0.3
	Instruct-GPTQ-Int4	Instruct-GPTQ-Int8	Chat-GPTQ	Instruct-INT4-GPTQ	Instruct-GPTQ-Q_8	Instruct-GPTQ-4bit
Ours	0.9890	0.9913	0.9933	0.9450	1.0000	0.9970

Results and Analysis. Table A7 shows that AttnDiff remains highly stable under GPTQ quantization across all evaluated families, with similarity scores close to 1.0 for most INT4/INT8 settings. The slightly lower similarity on the INT4-quantized Llama-3.1-8B variant indicates that more aggressive quantization can introduce stronger weight perturbations, but AttnDiff still preserves a high provenance signal overall.

F.8 Stress Testing LLMMAP under Text-level Attacks

We provide a focused stress test of the *text-based* provenance baseline LLMMAP [37]. Since LLMMAP operates on generated responses, it may be sensitive to post-hoc rewriting that preserves semantics but perturbs lexical and stylistic cues.

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Attack setup. We attack only the answers produced by each suspect model while leaving base model outputs unchanged. Using a local `WizardMath-7B-V1.0` model as the attacker, we rewrite each answer under three semantic-preserving transformations: **paraphrase** (synonym-based paraphrase), **rewrite** (stylistically distinct rewrite), and **style-transfer** (rewrite conditioned on a target writing style). After rewriting, we recompute LLMMAP embeddings and fingerprints and measure cosine similarity against the clean fingerprints.

Results and Analysis. Table A8 reports the cosine similarity of LLMMAP fingerprints before and after meaning-preserving paraphrase, rewrite, and style-transfer attacks. Performance varies widely across suspects: `CodeLlama-7b-hf` remains stable (clean 0.9535 and ≥ 0.9484 after attacks), while `llemma_7b` and `WizardMath-7B-V1.0` suffer substantial degradation (e.g., `llemma_7b`: $0.8886 \rightarrow 0.715$–0.743). The logit-watermark distilled model also exhibits a measurable drop ($0.9233 \rightarrow 0.885$–0.893). These results highlight a vulnerability of text-based provenance signals to post-hoc rewriting that preserves semantics but alters surface form, motivating complementary fingerprints derived from internal model dynamics.

Table A8: Cosine similarity of LLMMAP fingerprints before and after meaning-preserving paraphrasing, rewriting, and style-transfer attacks.

Model	Clean	Paraphrase	Rewrite	Style-transfer
CodeLlama-7b-hf	0.9535	0.9484	0.9484	0.9500
llemma_7b	0.8886	0.7174	0.7152	0.7433
llama-2-7b-logit-watermark-distill	0.9233	0.8934	0.8904	0.8851
WizardMath-7B-V1.0	0.6998	0.6147	0.6210	0.6330

G Broader Impacts

While ATTNDIFF provides a practical building block for model provenance and IP protection in the open-weight ecosystem, we acknowledge potential negative societal impacts. Specifically, inaccurate provenance attribution could lead to unwarranted copyright disputes or legal accusations against legitimate open-source developers, potentially chilling collaborative innovation. Mitigating these risks requires combining technical verification with human-in-the-loop auditing and legal frameworks that account for the statistical nature of fingerprinting signals.

H Model List

This section provides a comprehensive list of the models used across all experiments in the main text and appendix. We categorize these models by modification or attack setting (e.g., fine-tuning, model merging, pruning, distillation). For each category, we specify the base (victim) model used to establish the reference fingerprint and the corresponding suspect (derivative) models evaluated for provenance verification.

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Table A9: Models used in the main paper and appendix analyses. The evaluation spans fine-tuning, merging, pruning, preference optimization, distillation, and quantization across multiple families and scales.

Category	Type	Base (Victim)	Suspect(s) / Derivative(s)
Fine-tuning	Instruction	Llama-2-7B	Llama-2-finance-7b, Vicuna-1.5-7b, WizardMath-7b, Chinese-LLaMA-2-7b, CodeLLaMA-7b, Llemma-7b
Merge	Weight	Shisa-gamma-7b-v1, WizardMath-7b-1.1, Abel-7b-002	Evollm-jp-7b
	Dist./Behav.	Llama-2-7B, OpenLLaMA-2-7b, mpt-7b	Fusellm-7b
Pruning	Structured	Llama-2-7B	Sheared-llama-1.3b, Sheared-llama-1.3b-pruned, Sheared-llama-1.3b-sharegpt, Sheared-llama-2.7b, Sheared-llama-2.7b-pruned
	Unstructured	Llama-2-7B	Sparse-llama-2-7b, Wanda-llama-2-7b, GBLM-llama-2-7b
Ablation	Related	Llama-2-7B	CodeLlama-7b, Llama-2-finance-7B, Vicuna-7B-v1.5, Chinese-Llama-2-7B, WizardMath-7B-V1.0, llemma_7b, Sheared-LlaMA-1.3B, Sheared-LlaMA-1.3B-Pruned, Sheared-LlaMA-1.3B-ShareGPT, Sheared-LlaMA-2.7B, Sheared-LlaMA-2.7B-Pruned, Sheared-LlaMA-2.7B-ShareGPT, Sparse-llama-2-7b, Wanda-llama-2-7b, GBLM-llama-2-7b
	Unrelated	Llama-2-7B	Llama3-8B, mpt-7b, Qwen2.5-1.5B, Qwen2.5-3B, Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-Math-7B, gemma-2-2b, Gemma-7B-it, Yi-6B
Pilot	Discovery/Validation	Llama-2-7B	Llama-2-7B, CodeLlama-7b-hf, WizardMath-7B-V1.0, llemma_7b, Qwen2.5-7B
Pref. Opt.	PPO/DPO	Llama-2-7B	llama-2-7b-ppo-v0.1-reward[1], llama-2-7b-ppo-v0.1-policy[1], tulu-2-dpo-7b[2], llama-2-7b-dpo[3]
Cross-Family/Scale	Qwen2.5	Qwen2.5-7B	Qwen2.5-Coder-1.5B[4], Qwen2.5-Math-1.5B[5], Qwen2.5-1.5B-Instruct[6]
		Qwen2.5-14B	Qwen2.5-14B-Instruct[7], oxy-1-small[8], Qwen2.5-14B-Gutenberg-Instruct-Slerpeno[9]
	Gemma-2	gemma-2-2b	gemma-2-2b-neogenes-ita[10], gemma-2-baku-2b[11], gemma-2-2b-merged[12]
	Mistral	Mistral-7B-v0.3	AQUA-7B[13], spellcheck-mistral-7b[14], Mistral-7B-Instruct-demi-merge-v0.3-7B[15]
Distill	Reasoning	Llama-3.1-8B, Qwen2.5-7B, Qwen2.5-14B	Llama-3.1-8B-Instruct-Open-R1-Distill[16], Qwen2.5-7B-Open-R1-Distill[17], DeepSeek-R1-Distill-Qwen-14B[18]
	Logit-based	Llama-2-7B	llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2[19]
MoE	Mixtral	Mixtral-8x7B	Instruct_Mixtral-8x7B-v0.1_Dolly15K[20], Nous-Hermes-2-Mixtral-8x7B-DPO[21], openbuddy-mixtral-8x7b-v15.4[22]
Quantization	GPTQ	Qwen2.5-7B	Qwen2.5-7B-Instruct-GPTQ-Int8[23], Qwen2.5-7B-Instruct-GPTQ-Int4[24]
		Llama-2-7B	Llama-2-7B-Chat-GPTQ[25]
		Llama-3.1-8B	Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8[26], LLaMA-3.1-8B-Instruct-INT4-GPTQ[27]
		Mistral-7B-v0.3	Mistral-7B-Instruct-v0.3-GPTQ-4bit[28]

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Table A10: Repository URLs corresponding to the superscript IDs in Table A9. All checkpoints are publicly available on Hugging Face.

ID	Repository URL
1	https://huggingface.co/renyiyu/llama-2-7b-ppo-lora-v0.1
2	https://huggingface.co/allenai/tulu-2-dpo-7b
3	https://huggingface.co/mncai/llama2-7b-dpo-v1
4	https://huggingface.co/Qwen/Qwen2.5-Coder-1.5B
5	https://huggingface.co/Qwen/Qwen2.5-Math-1.5B
6	https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct
7	https://huggingface.co/Qwen/Qwen2.5-14B-Instruct
8	https://huggingface.co/oxyapi/oxy-1-small
9	https://huggingface.co/v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno
10	https://huggingface.co/anakin87/gemma-2-2b-neogenesis-ita
11	https://huggingface.co/rinna/gemma-2-baku-2b
12	https://huggingface.co/vonjack/gemma2-2b-merged
13	https://huggingface.co/KurmaAI/AQUA-7B
14	https://huggingface.co/openfoodfacts/spellcheck-mistral-7b
15	https://huggingface.co/grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B
16	https://huggingface.co/asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill
17	https://huggingface.co/erickrus/Qwen2.5-7B-Open-R1-Distill
18	https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Qwen-14B
19	https://huggingface.co/cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2
20	https://huggingface.co/Brillibits/Instruct_Mixtral-8x7B-v0.1_Dolly15K
21	https://huggingface.co/NousResearch/Nous-Hermes-2-Mixtral-8x7B-DPO
22	https://huggingface.co/openbuddy/openbuddy-mixtral-8x7b-v15.4
23	https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8
24	https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4
25	https://huggingface.co/TheBloke/Llama-2-7B-Chat-GPTQ
26	https://huggingface.co/iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8
27	https://huggingface.co/DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ
28	https://huggingface.co/RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit

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